

Mémoire M2

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Abstract

In this paper we discuss various results relating to Central Limit Theorems for Point Processes. We start by giving a quick review of the cumulant method for point processes with the usual hypotheses and give a self-contained proof of the famous Marcinkiewicz theorem obtained by combining several existing proofs. Secondly and most importantly, we study a particular class of hyper-uniform point processes for which their smooth statistics follow a generalised central theorem: the fluctuations converge to a stable distribution. The class that we study are α -hyper-uniform point processes perturbed by a β -stable perturbation in the case $d = 1$ with $\alpha > \beta$, $\beta \in]1, 2[$. We will see why this is the class that we study and then a proof of the convergence to the stable distribution by studying its characteristic function. This result is an extension of the work done in [Gabriel Mastrilli]. A lot of the proofs in our paper are very similar or at least inspired to those found in [Gabriel Mastrilli].

Keywords— Central limit theorem, Generalized central limit theorem, Point process, Hyper-uniform point process, Marcinkiewicz's theorem, smooth linear statistics

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Notations and definitions

General

In the whole paper, $d \in \mathbb{N}^*$ and will denote the dimension of the space we are working with, $\mathbb{R}^d$. To lighten notation in our calculations, we will denote the vector $\mathbf{a} = (x_1, \dots, x_n)$ by $[x]_1^n$.

We denote by $\mathcal{B}(\mathbb{R}^d)$ the set of Borel sets of $\mathbb{R}^d$ and we denote by $\mathcal{B}_c(\mathbb{R}^d)$ the set of bounded Borel sets.

For a function $f \in L^1(\mathbb{R}^d)$, we denote its Fourier transform by $\mathcal{F}[f]$, defined by:

$$\mathcal{F}[f](k) := \frac{1}{(2\pi)^{\frac{d}{2}}} \int_{\mathbb{R}^d} f(x) e^{ik \cdot x} dx$$

Point processes

We denote by $\overline{\mathbb{M}}(\mathbb{R}^d)$ the set of locally finite measures on $\mathbb{R}^d$ and $\overline{\mathcal{M}}(\mathbb{R}^d)$ the smallest σ -algebra which makes the mappings $\mu \mapsto \mu(B)$ for $B \in \mathcal{B}(\mathbb{R}^d)$.

We denote by $\mathbb{M}(\mathbb{R}^d)$ the set of counting measures: the measures of $\overline{\mathbb{M}}(\mathbb{R}^d)$ which only take integers or ∞ as values. Similarly, we define $\mathcal{M}(\mathbb{R}^d)$ as the σ -algebra defined by the mappings $\mu \mapsto \mu(B)$ for $B \in \mathcal{B}(\mathbb{R}^d)$ but for $\mu \in \mathbb{M}(\mathbb{R}^d)$ this time.

1 General context and overview of the paper's content

This paper attempts to provide a few results related to Central limit theorems for point processes. In the first section, we discuss the cumulant method for proving Central Limit Theorems for Point Processes with typical assumptions. The methods hinges on the central theorem of Marcinkiewicz [Add ref]. However, it is hard to find a proof of this result even though it is often referred to. We wanted to present here a fully self-contained proof of that result for future references since it is a very useful result. The proof is based on Cramér's theorem, of which we also give a full proof.

After that, we give a general (but not new) result regarding smooth statistics of Point processes giving us a Central Limit theorem with typical assumptions. This kind of result can be found in [Add ref].

In the following section, we focus on hyper-uniform point processes and specifically perturbed hyper-uniform point processes. The new result of this part is an extension of the result found in [add ref of precise spot in Gabriel's paper]. In [ref again] the case studied is the perturbation of the stationnarized $\mathbb{Z}^d + U$ which is hyper uniform (informally, its hyper-uniform exponent is ∞). The results given by Gabriel cover various central limit theorems depending on two parameters: the dimension d and the parameters β of the β -stable perturbation. In his case, he finds that there is a convergence to a non-gaussian limit for $d = 1$ and $\beta \in [1, 2[$.

Here we look at the same case: $d = 1$ and $\beta \in]1, 2[$. The novelty of our result is that we add an additional parameter by starting from an arbitrary α -hyper-uniform (not obtained by perturbation, which is the typical way of obtaining α -hyper-uniform point processes for a chosen α) point process which is stationary with $\alpha > 1$ and we require $\alpha > \beta$. Following these conditions, we obtain that the fluctuations of a smooth statistic of the point process converge to a β -stable distribution. The conditions mentioned previously seem necessary for there to be a non-gaussian limit for reasons outlined in 4.3.2. On top of these mandatory assumptions, our calculation seem to indicate there is an additional condition: $\alpha > \frac{3\beta-2}{\beta}$ which is slightly more constraining than simply $\alpha > \beta$ for $\beta \in]1, 2[$ (see figure 1).

2 Some generalities and tools for point processes

2.1 General framework

In this section we give some definitions and set some conventions for the rest of the paper. For example we precisely define what we mean by "Point Process" seen as a random measure, and review a few key objects in relation with point processes such as Mean Measure and so on. We also justify in our cases operations such as enumerations of the points of the realizations of the random measures.

2.1.1 Random measures and Point Processes

Most of the definitions that will follow can be set in a very abstract topological space that is Locally Compact Second countable and Hausdorff (commonly abbreviated into l.c.s.h). However our scope will be limited to working with $\mathbb{R}^d$ so we will keep our definitions within that scope.

Unless specified otherwise we will be working in the usual euclidean $\mathbb{R}^d$ space endowed with its usual topology $\mathcal{B}(\mathbb{R}^d)$ (which is obviously l.c.s.h.).

Definition 2.1 (Random Measure). *A random measure Φ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is a measurable mapping $\Phi : (\Omega, \mathcal{F}) \rightarrow (\overline{\mathbb{M}}(\mathbb{R}^d), \overline{\mathcal{M}}(\mathbb{R}^d))$*

General measures are an object that is too vast and not suited for us so we only look at countable measures:

Definition 2.2 (Countable Measure). *A measure μ is said to be countable if $\forall B \in \mathcal{B}_c(\mathbb{R}^d), \mu(B) \in \mathbb{N}$*

Naturally, we look at:

Definition 2.3 (Point Process). *A Point Process is a measurable mapping $\Phi : (\Omega, \mathcal{F}) \rightarrow (\overline{\mathbb{M}}(\mathbb{R}^d), \overline{\mathcal{M}}(\mathbb{R}^d))$*

This framework is very flexible and powerful and it is generally easier to look at Point Processes as random measures rather than random sets, although that point of view might be more intuitive.

In most cases, we will be working with simple Point Processes:

Definition 2.4 (Simple Point Process). *A Point Process Φ is said to be simple if $\forall x \in \mathbb{R}^d, \mathbb{P}\text{-a.s. } \Phi(\{x\}) \leq 1$ (That is to say that there is almost surely never two points in the same place).*

An operation that seems intuitively straight forward is the enumeration of the points of a realization of a Point Process. However intuitive it may seem, it is not so obvious in general so we show it rigorously here (we use the proof found in [Ref Random Measures, Point Processes, and Stochastic Geometry]).

We first show this technical lemma stating that any counting measure μ has a diffuse component that is the null measure, so it is purely atomic (this is done without any randomness at this point). Then we will find a measurable way to enumerate the atoms of any countable measure (it is critical that the space be appropriate for that).

For a counting measure μ , we denote $\mu = \mu_a + \mu_d$ where μ_a is the restriction of μ to $A = \{x \in \mathbb{R}^d, \mu(\{x\}) > 0\}$.

Lemma 2.1. *Counting measures on $\mathbb{R}^d$ are purely atomic.*

Proof. Let μ be a countable measure on $\mathbb{R}^d$. First we prove that for every $x \in \mathbb{R}^d$ there exists a neighborhood U_x such that $\mu_d(U_x) = 0$ and then we will extend that to show that μ_d is the null measure.

Assume by contradiction that all neighborhoods U of x , $\mu_d(U) \geq 1$. Then let $(U_n)_{n \geq 0}$ be a sequence of decreasing relatively compact neighborhoods of x such that: $\bigcap_{n \geq 0} U_n = \{x\}$. By continuity from above of measures, we have that $\mu_d(\{x\}) = \lim_{n \rightarrow \infty} \mu_d(U_n) \geq 1$ which is a contradiction.

Now we extend the result everywhere: for any compact set K from the covering $\bigcup_{x \in K} U_x$ we can extract a finite covering $\bigcup_{i=0, \dots, N} U_{x_i}$ (where U_x is the neighborhood obtained from the previous reasoning). Therefore, $\mu_d(K) \leq \sum_{i=0}^N \mu_d(U_{x_i}) = 0$. Then we can

take a sequence $(K_n)_{n \geq 0}$ of compact sets which increase to $\mathbb{R}^d$ and each of are such that $\mu_d(K_n) = 0$. Thus :

$$\mu_d(\mathbb{R}^d) = \lim_{n \rightarrow \infty} \uparrow \mu_d(K_n) = 0$$

□

With this lemma, we can then show that we can enumerate the atoms of counting measures in a measurable way:

Proposition 2.1. *Any measure $\mu \in \mathbb{M}(\mathbb{R}^d)$ can be written:*

$$\mu = \sum_{i \in I} a_i \delta_{x_i}$$

Where $I \in \overline{\mathbb{N}}$ and $(x_i)_{i \in I}$ is a sequence of points of $\mathbb{R}^d$ and $(a_i)_{i \in I}$ are strictly positive integers. This decomposition is unique up to a permutation of the order of the terms.

Proof. $\mathbb{R}^d$ has a countable partition of relatively compact sets $B_0, B_1, \dots$ and since μ is locally finite, then $\mu(B_i) < \infty$ for all $i \in \mathbb{N}$. Therefore, the number of $x \in B_i$ such that $\mu(\{x\}) \geq 1$ is finite (otherwise $\mu(B_i) = \infty$ which contradicts local finiteness). We denote the number of such x by N_i . We can index them in the following way: $(x_{i,1}, \dots, x_{i,N_i})$ are the atoms of μ in B_i and we denote $\mu(\{x_{i,j}\})$ by $(a_{i,j})_{1 \leq j \leq N_i}$. As a result, we have:

$$\mu = \sum_{i \geq 0} \sum_{j=1}^{N_i} a_{i,j} \delta_{x_{i,j}}$$

And we can "unravel" the indexing to obtain the desired form:

$$\mu = \sum_{i \in I} a_i \delta_{x_i}$$

□

2.1.2 Mean measure and Laplace transforms for Point Processes

A very useful tool to describe the structure of a point process is its Mean Measure:

Definition 2.5 (Mean Measure). *Let Φ be a point process on $\mathbb{R}^d$. Its mean measure M_Φ is defined on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ by:*

$$M_\Phi(B) = \mathbb{E}[\Phi(B)], \quad B \in \mathcal{B}(\mathbb{R}^d)$$

A particularly useful use case of the mean measure is the Campbell averaging formula:

Theorem 2.2 (Campbell's averaging formula). *Let Φ be a point process on $\mathbb{R}^d$ with its mean measure M_Φ .*

Then for all positive measurable functions f or $f \in L^1(\mathbb{R}^d)$ (with real or complex values) we have the following formula:

$$\mathbb{E} \left[\int_{\mathbb{R}^d} f d\Phi \right] = \int_{\mathbb{R}^d} f dM_\Phi$$

Proof. The proof is classical: we start by proving it for indicatrix function, then we construct positive measurable functions as an increasing limit of indicatrix functions and so on... And for L^1 functions we treat the positive and negative part separately and apply the same result. $\square$

Another tool which can be useful to characterize point processes is its Laplace transform:

Definition 2.6 (Laplace Transform of a point process). *Let Φ be a point process on $\mathbb{R}^d$. Its Laplace transform is a functional defined on all positive measurable functions $f : \mathbb{R}^d \rightarrow \overline{\mathbb{R}}_+$ by:*

$$\mathcal{L}_\Phi(f) = \mathbb{E} \left[e^{-\int_{\mathbb{R}^d} f d\Phi} \right]$$

It is particularly useful because it completely characterizes the law of a point process and it is pretty convenient to manipulate.

Lastly we define the generating function of a point process:

Definition 2.7 (Generating function of a point process). *For $f : \mathbb{R} \rightarrow \mathbb{C}$ such that $|f| \leq 1$ we define the generating function of a point process Φ by:*

$$\mathcal{G}_\Phi(f) = \mathbb{E} \left[\prod_{x \in \Phi} f(x) \right]$$

2.1.3 Independant displacements of a Point Process

In our next sections we will deal with point processes obtained by an independant displacement of all points by an independant perturbation. The idea is pretty intuitive but we give a rigorous definition here:

Definition 2.8. *Let Φ be a point process on $\mathbb{R}^d$ and p a probability kernel from $\mathbb{R}^d$ to itself. We consider a measurable enumeration $X_0, X_1, \dots$ of the points of Φ and $(Y_k)_{k \in \mathbb{N}}$ an independant sequence such that:*

$$\mathbb{P}(Y_k \in \cdot \mid \Phi) = p(X_k, \cdot), \quad \forall k \in \mathbb{N}$$

Then the new point process:

$$\tilde{\Phi} := \sum_{k \in \mathbb{N}} \delta_{Y_k}$$

is called an independant displacement of the point process Φ .

In the particular case where the displacements are i.i.d and not dependant on the position of the X_k we can rewrite it in the following way:

$$\mathbb{P}(Y_k \in \cdot \mid \Phi) = p(\cdot + X_k)$$

Or equivalently:

$$P(Y_k - X_k \in \cdot \mid \Phi) = p(\cdot)$$

This is the same as having an independant i.i.d perturbation $(\xi_k)_{k \in \mathbb{N}}$ and having the new point process be:

$$\tilde{\Phi} = \sum_{k \in \mathbb{N}} \delta_{X_k + \xi_k}$$

2.1.4 Palm theory and stationary Point Processes

In some cases it is useful to "observe" a point process not from an arbitrary point in space but from the perspective of a given point of the point process. The way to do this is by using Palm theory which gives us the new probability distribution under that new observation. By thinking about Poisson processes it doesn't appear very useful because the Palm distribution observed from x is simply the same distribution with an added atom on x : $\Phi_x \stackrel{d}{=} \Phi + \delta_x$ where Φ_x is the palm distribution associated with Φ observed from point x . However, if one looks at, for example, a Matérn hard-core process obviously changes a lot compared to when you look at it from an arbitrary point in space, or an arbitrary point of Φ : in the first case you can have a point in the neighborhood of the observation point, and in the second you cannot. Intuitively, the Palm distribution Φ_x can be understood as "the conditional probability of Φ given that $x \in \Phi$ ". This is a little simplistic to be exactly that but the idea is there.

We won't be using Palm theory here. The reason I mention it is because to make it work with the property of "stationarity" it is useful to introduce a "shift operator" or "probability flow" which operates directly on the probability space Ω so that for $x \in \mathbb{R}^d$, we have $\theta_x \mu(B) := \mu(B+x)$, $\forall B \in \mathcal{B}(\mathbb{R}^d)$. As a result, this gives a definition of stationarity using the shift operator:

Definition 2.9 (stationarity for point processes with shift operator). *We say that a point process Φ on $\mathbb{R}^d$ is stationary if:*

$$\forall x \in \mathbb{R}^d, \mathbb{P}_{\theta_x \circ \Phi} = \mathbb{P}_\Phi$$

However if we don't use Palm theory, we can use a simpler definition:

Definition 2.10 (stationary for point process without shift operator). *We say that a point process Φ on $\mathbb{R}^d$ is stationary if :*

$$\Phi =: (X_k)_{k \geq 0} \stackrel{d}{=} (X_k + x)_{k \geq 0} := \Phi + x$$

One important property of stationary point processes which we will use throughout the computations is the following:

Proposition 2.2. *The mean measure of a stationary point process Φ is proportionnal to the Lebesgue measure ℓ :*

$$\exists \lambda \geq 0, M_\Phi = \lambda \ell$$

Proof. Let Φ be a stationary point process on $\mathbb{R}^d$ and f a positive measurable function. By Campbell's formula:

$$\mathbb{E} \left[\int_{\mathbb{R}^d} f(x) d\Phi(x) \right] = \int_{\mathbb{R}^d} f(x) dM_\Phi(x)$$

By stationarity, for any $y \in \mathbb{R}^d$ we also have:

$$\mathbb{E} \left[\int_{\mathbb{R}^d} f(x) d\Phi(x) \right] = \mathbb{E} \left[\int_{\mathbb{R}^d} f(x+y) d\Phi(x) \right]$$

Which is equal, by Campbell's formula again, to $\int_{\mathbb{R}^d} f(x+y) dM_\Phi(x) = \int_{\mathbb{R}^d} f(x) dM_\Phi(x-y)$. Thus we have:

$$\int_{\mathbb{R}^d} f(x) dM_\Phi(x) = \int_{\mathbb{R}^d} f(x) dM_\Phi(x-y)$$

Which means that M_Φ is translation invariant, which means that it is a multiple of the Lebesgue measure (by Haar's theorem for example). $\square$

2.2 Tools for Point Processes

In this section we will introduce the main tools we will be using to prove our results on point processes: the moment and cumulant measures and their factorial version, the characteristic function and its series expansions.

2.2.1 The Characteristic function

Definition 2.11 (Characteristic function of a Point Process). *For a point process Φ and f a positive and measurable function (or in $L^1(M_\Phi, \mathbb{R}^d)$) we define:*

$$\Psi_\Phi(f) := \mathbb{E} \left[\exp \left(i \int_{\mathbb{R}^d} f d\Phi \right) \right]$$

Remark. Ψ_Φ is continuous (and even Lipschitz), which can be seen using the inequality $|e^{ix} - e^{iy}| \leq |x - y|$ for $x, y \in \mathbb{R}$.

2.2.2 Moment and Cumulant measures

Definition 2.12 (Moment measure). *For a point process Φ we define its n -th order moment measure M_{Φ^n} by:*

$$M_{\Phi^n}(B_1 \times \cdots \times B_n) := \mathbb{E} [\Phi(B_1) \cdots \Phi(B_n)]$$

Definition 2.13 (Factorial moment measure). *For a point process Φ we define its n -th order factorial moment measure $M_{\Phi^{(n)}}$:*

$$M_{\Phi^{(n)}}(B_1 \times \cdots \times B_n) = \sum_{j=1}^n (-1)^{n-j} \sum_{\pi \in \binom{[n]}{j}} \mathbb{E} \left[\prod_{i \in \pi} \Phi(B_i) \right]$$

Where " $\pi \in \binom{[n]}{j}$ " denotes a subset of $\{1, \dots, n\}$ with exactly j elements.

[To do: Add some explanations about moment/factorial/cumulant measures there]

2.2.3 Series expansion (Maybe replace this with the "Janossy expansion" + Janossy measure etc.)

Another tool which we will use for proving our main result will be the Janossy measures and the correspondent serie expansion (which we will name "Janossy expansion" here, but is simply a rewriting of the generating function using the Janossy measures).

Definition 2.14 (Janossy measure). *Let Φ be a **finite** point process on $\mathbb{R}^d$. For any $n \in \mathbb{N}^*$, the Janossy measure of order n J_n is defined by:*

$$J_n(B) := n! p_n \Pi_n, \quad B \in \mathcal{B}(\mathbb{R}^d)$$

Where $p_n = \mathbb{P}(\Phi(\mathbb{R}^d) = n)$ for $n \in \mathbb{N}$ and :

$$\Pi_n(B) := \frac{1}{n!} \sum_{\sigma} \mathbb{P}((X_{\sigma(1)}(\Phi), \dots, X_{\sigma(n)}(\Phi)) \in B | \Phi(\mathbb{R}^d) = n), \quad B \in \mathcal{B}(\mathbb{R}^d)$$

where the summation is over all permutations of $\{1, \dots, n\}$. The summation makes Π_n symmetric!

Remark. The Janossy measures along with p_n and Π_n provide an intuitive construction of a **finite** point process: generate its number of points according to the distribution $(p_n)_{n \in \mathbb{N}}$, then the spatial distribution of the n points is given by Π_n .

Now that we have the Janossy measures defined, we have the following:

Proposition 2.3 ("Janossy expansion"). For Φ a finite point process on $\mathbb{R}^d$ and for $f : \mathbb{R} \rightarrow \mathbb{C}$ such that $|f| \leq 1$, we have the following result:

$$\mathcal{G}(f) = p_0 + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{(\mathbb{R}^d)^n} \left(\prod_{i=1}^n f(x_i) \right) J_n(d[x]_1^n)$$

In this paper we will be using it in the case of the characteristic function:

$$\mathbb{E}[e^{-it \sum_{x \in \Phi} f(x)}] = p_0 + \sum_{n \geq 1} p_n \int_{\mathbb{R}^d} e^{-it \sum_{i=1}^n f(x)} \Pi_n(d[x]_1^n)$$

3 The cumulant method for the Central Limit Theorem

Many methods exist to prove central limit theorems in a variety of cases. One of those is the cumulant method. It is often used in the case of point processes, among others.

The cumulant method consists in showing the convergence of the cumulants of the statistic being studied towards those of the normal law. This convergence implies the convergence in distribution towards the Gaussian law (since we can identify the cumulants of the Gaussian characteristic function).

Proving the convergence of the cumulants amounts to finding bounds on them, and with the expression of the cumulants, there are many combinatorial tools available for that [add ref maybe Gabriel Mastrilli?]. However, the main upside of the cumulant method for the CLT lies in the central result that is Marcinkiewicz's theorem [ref], which states that the only distribution [hypothesis on the moments? maybe just finite variance?] whose cumulant expansion only has finitely many terms (namely, 2), is the Gaussian distribution.

As a result, it is sufficient to show bounds for cumulants of order $p \geq n$ for some $n \geq 3$.

During my research, I have found many papers quoting the important theorem of Marcinkiewicz, but the actual proof was hard to find. For clarity purposes, in this section, we provide a full proof of Marcinkiewicz's theorem [ref: cf. my emails] for this result. It uses Cramér's theorem [ref], for which we also give a self-contained proof.

3.1 Cramér's theorem and lemmas

3.1.1 Lemma 1

This lemma is a common complex analysis result, but we include it for the sake of completeness.

Lemma 3.1. *Let f be an entire function which vanishes nowhere. Then f has a well-defined logarithm g , which is also an entire function. In other words: there exists g an entire function such that $f = e^g$*

Proof. The exponential map $g : \mathbb{C} \rightarrow \mathbb{C}^*$ is surjective. So there exists $z_0 \in \mathbb{C}$ such that $f(z_0) = \exp(z_0)$ since f vanishes nowhere. As a ratio of an entire function and an entire function with no zeroes, the function $\frac{f'}{f}$ is also entire. As a result, it admits an entire function g as a primitive which we can construct explicitly:

$$g(z) := z_0 + \int_{[0,z]} \frac{f'(u)}{f(u)} du$$

We will now show that g is the desired function.

Let $v := \exp(g)$.

$$v' = g' \exp(g) = \frac{f'}{f} v$$

So :

$$\left(\frac{v}{f}\right)' = \frac{v'f - f'v}{f^2} = 0$$

So the ratio $\frac{v}{f}$ is a constant. Since we have $v(0) = \exp(z_0) = f(0)$, we have shown that $v = f$, so $f = e^g$ $\square$

3.1.2 Lemma 2

Lemma 3.2. *Let f be an entire function, and P a polynomial such that :*

$$\forall z \in \mathbb{C}, \operatorname{Re}(f(z)) \leq |P(z)|$$

Then f is a polynomial with degree at most equal to that of P .

Remark. *This second technical lemma is key to obtain the fact that the characteristic function is of the form $\exp(P)$ with P a polynomial of order at most 2.*

Proof. By assumption, we can write $\forall z \in \mathbb{C}, f(z) = \sum_{n \geq 0} a_n z^n$. We denote $P(z) = \sum_{n=0}^d b_n z^n$ with d the degree of P . The Cauchy formulas give:

$$\forall n \in \mathbb{N}^* \quad a_n r^n = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) e^{-in\theta} d\theta \quad (1a)$$

$$0 = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) e^{in\theta} d\theta \quad (1b)$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) d\theta \quad (1c)$$

Adding (1a) and the conjugate of (1b), and taking the real part of (1c) gives us :

$$\forall n \in \mathbb{N}^* \quad |a_n| r^n = \frac{1}{\pi} \int_0^\pi \operatorname{Re}(f(re^{i\theta})) e^{-in\theta} d\theta \quad (2a)$$

$$2 \operatorname{Re}(a_0) = \frac{1}{\pi} \int_0^\pi \operatorname{Re}(f(re^{i\theta})) d\theta \quad (2b)$$

Adding (2a) and (2b) gives :

$$\begin{aligned} \forall n \in \mathbb{N}^*, |a_n|r^n + 2 \operatorname{Re}(a_0) &\leq \frac{1}{\pi} \int_0^{2\pi} (|\operatorname{Re}(f(re^{i\theta}))| + \operatorname{Re}(f(re^{i\theta}))) d\theta \\ &= 4 \max_{|z|=r} (\operatorname{Re}(f(z)), 0) \end{aligned}$$

$$\text{Which by hypothesis gives } \leq 4 \max_{|z|=r} (|P(z)|, 0)$$

Since the dominant term of the right hand term is r^d , for this inequality to be satisfied for every n (in particular as $r \rightarrow \infty$), it is necessary that for $n > d$ we have $a_n = 0$ which is the result we wanted. $\square$

3.1.3 Lemma 3

This lemma gives a characterization of the normal law which will be useful later on.

Lemma 3.3. *Let X be a real random variable such that there exists $\eta > 0$ such that $f(\eta) := \mathbb{E}[e^{\eta^2 X^2}] < \infty$ and such that its characteristic function φ_X vanishes nowhere. Then X is necessarily normal.*

Proof. Let $\eta > 0$, $\zeta \in \mathbb{C}$ and $x \in \mathbb{R}$. We use the following trick $|\zeta x| \leq |\zeta \eta^{-1} x| \leq (\eta^2 x^2 + \eta^{-2} |\zeta|)$ to get, by Jensen and the fact that the exponential map is increasing over the reals:

$$|\mathbb{E}[e^{i\zeta X}]| \leq \mathbb{E}[|e^{i\zeta X}|] \leq \mathbb{E}[e^{|\zeta X|}] \leq \mathbb{E}[e^{\eta^2 X^2 + \eta^{-2} |\zeta|}]$$

The assumption gives us exactly the necessary bound to conclude that φ_X is holomorphic. Indeed, it suffices to work for $|\zeta| < R$ for arbitrary values of R .

By our assumption, Φ vanishes nowhere and is holomorphic by the previous result. By lemma 3.1, we can define g an entire function such that $\varphi_X = \exp(g)$. As a result of the previous inequality, we have:

$$\forall \zeta \in \mathbb{C} \quad |\exp(g(\zeta))| \leq e^{\eta^{-2} |\zeta|^2} f(\eta) \implies \operatorname{Re}(g(\zeta)) \leq \eta^{-2} |\zeta|^2 + \ln(f(\eta))$$

By lemma 3.2, g is a polynomial of degree at most 2:

$$g(\zeta) = a_0 + ia_1 \zeta + \frac{a_2}{2} \zeta^2$$

Since $\varphi_X(0) = 1$, $a_0 = 0$. Moreover, since for all $k \in \mathbb{N}$ we have $\mathbb{E}[X^k] = (-i)^k \varphi_X^{(k)}(0)$, in particular we have $a_1 = \mathbb{E}[X] =: \mu$ and $a_2 = -\operatorname{Var}(X) =: -\sigma^2$.

So $\varphi_X(\zeta) = \exp(i\mu\zeta - \frac{1}{2}\sigma^2\zeta^2)$, i.e. X follows a normal distribution $\mathcal{N}(\mu, \sigma^2)$ $\square$

3.1.4 Lemma 4

In this final lemma, we will write f in term of the probability of X . This final element will allow us to prove Cramér's theorem.

Lemma 3.4. *If X has no atoms, the function f introduced earlier : $f(\eta) =: \mathbb{E}[e^{\eta^2 X^2}]$ can be written as follows :*

$$f(\eta) = 1 + 2 \int_0^\infty x \eta^2 e^{x^2 \eta^2} \mathbb{P}(|X| \geq x) dx \quad (3)$$

Remark. This is simply a slightly different version of the usual trick :

$$\mathbb{E}[X] = \int_0^\infty x \mathbb{P}(X \geq x) dx$$

Proof. We denote $\rho(x)$ the density of X . First we split the integral and then we will do an integration by part using respectively $\mathbb{P}(X \leq \cdot)$ and $-\mathbb{P}(X \geq \cdot)$ as primitives of ρ :

$$\begin{aligned} f(\eta) &= \int_{-\infty}^0 e^{\eta^2 x^2} \rho(x) dx + \int_0^\infty e^{\eta^2 x^2} \rho(x) dx \\ &= \left[e^{\eta^2 x^2} \mathbb{P}(X \leq x) \right]_{-\infty}^0 - \int_{-\infty}^0 2x\eta^2 e^{\eta^2 x^2} \mathbb{P}(X \leq x) dx \\ &\quad + \left[-e^{\eta^2 x^2} \mathbb{P}(X \geq x) \right]_0^\infty + \int_0^\infty 2x\eta^2 e^{\eta^2 x^2} \mathbb{P}(X \geq x) dx \\ &= \mathbb{P}(X \leq 0) + \int_0^\infty 2x\eta^2 e^{\eta^2 x^2} \mathbb{P}(X \leq -x) dx \\ &\quad + \mathbb{P}(X \geq 0) + \int_0^\infty 2x\eta^2 e^{\eta^2 x^2} \mathbb{P}(X \geq x) dx \\ &= 1 + 2 \int_0^\infty x\eta^2 e^{\eta^2 x^2} \mathbb{P}(|X| \geq x) dx \end{aligned}$$

□

3.1.5 Cramér's theorem

We are finally ready to prove Cramér's theorem :

Theorem 3.5. (Cramér's theorem) Let X_1 and X_2 be two independant real random variables. If $X_1 + X_2$ follow a Gaussian distribution, then both X_1 and X_2 are Gaussians (possibly degenerate).

Remark. This has been compared to propagation of chaos in dynamical systems, where a system can asymptotically approach chaos, but can not be truly chaotic if it wasn't to begin with: a sum of random variables (correctly renormalized) can converge to a Gaussian distribution but the convergence is asymptotical, and the sum never follows a truly Gaussian law, unless the variables were already gaussians.

Proof. There are two cases to deal with:

- If $\text{Var}(X_1 + X_2) = 0$, then by independence, we have $\text{Var}(X_1) = \text{Var}(X_2) = 0$. So both X_1 and X_2 are constants, also known as degenerate Gaussian laws.
- Let us then suppose that $\text{Var}(X_1 + X_2) > 0$

X_1 and X_2 can't both have atoms, otherwise we have (Let a_1 and a_2 be atoms of X_1 and X_2 respectively):

$$\begin{aligned} 0 < \mathbb{P}(X_1 = a_1) \mathbb{P}(X_2 = a_2) &= \mathbb{P}(X_1 = a_1, X_2 = a_2) \\ &\leq \mathbb{P}(X_1 + X_2 = a_1 + a_2) = 0 \end{aligned}$$

Since $X_1 + X_2$ is a Gaussian, which has no atoms (we already took care of the case where it is degenerate).

Let us then suppose that X_2 has no atoms. We can then split the distribution of X_2 : let $\mu \in \mathbb{R}$ such that $\mathbb{P}(X_2 \leq \mu) = \mathbb{P}(X_2 \geq \mu) = \frac{1}{2}$.

Let $x > 0$.

$$\begin{aligned}\mathbb{P}(|X_1| \geq x) &= (\mathbb{P}(X_1 \geq x) + \mathbb{P}(X_1 \leq -x)) \\ \text{By independance:} &= 2(\mathbb{P}(X_1 \geq x, X_2 \geq \mu) + \mathbb{P}(X_1 \leq -x, X_2 \leq \mu)) \\ &\leq 2(\mathbb{P}(X_1 + X_2 \geq x + \mu) + \mathbb{P}(X_1 + X_2 \leq -x + \mu)) \\ &= 2\mathbb{P}(|X_1 + X_2 - \mu| \geq x)\end{aligned}$$

By lemma 3.4, we have :

$$\begin{aligned}f_{X_1}(\eta) &= 1 + 2 \int_0^\infty x \eta^2 e^{\eta^2 x^2} \mathbb{P}(|X_1| \geq x) dx \\ \text{By the previous result} &\leq 2 + 4 \int_0^\infty x \eta^2 e^{\eta^2 x^2} \mathbb{P}(|X_1 + X_2 - \mu| \geq x) dx \\ &= 2f_{X_1 + X_2 - \mu}(\eta) < \infty\end{aligned}$$

Since $X_1 + X_2 - \mu$ follows a normal distribution.

Finally, since we can factorize the characteristic function $\varphi_{X_1 + X_2} = \varphi_{X_1} \varphi_{X_2}$ and since $\varphi_{X_1 + X_2}$ never vanishes, neither does φ_{X_1} . Then by lemma 3.3 $X_1 \sim \mathcal{N}$.

If X_1 is a constant, then $X_2 = (X_1 + X_2) - X_1$ which is Gaussian. Otherwise, we can apply the same proof for X_2 since in this case X_1 has no atoms either.

□

3.2 Marcinkiewicz's theorem

Finally equipped with Cramér's theorem, we can prove Marcinkiewicz's theorem which states:

Theorem 3.6. (Marcinkiewicz's theorem) *let X be a real random variable. If for all $s \in \mathbb{C}$ such that $|s| < R$ for some $R > 0$ we have*

$$\varphi_X(s) = \exp Q(s)$$

where Q is a polynomial, then Q is of degree at most 2 and X follows a Gaussian distribution.

Remark. *This is the crux of the cumulant method.*

Remark. *Thanks to Cramér's theorem, we will be able to use some symmetry arguments which will cause some cancelations in the computations.*

Proof. Let Q be a polynomial such that $\forall |s| < R$, $\varphi_X(s) = \exp Q(s)$ for some fixed $R > 0$. This equation defines the extension of φ_X to all of $\mathbb{C}$. So the equation $\varphi_X(s) = \mathbb{E}[e^{isX}]$ remains true on all of $\mathbb{C}$.

By Cramér's theorem, if $\varphi_X(s)\varphi_X(-s)$ is the characteristic function of a Gaussian distribution, then φ_X also is. The proof will consist in bounding $\varphi_X(s)\varphi_X(-s)$ from below and above for a precise $s \in \mathbb{C}$, which will force the degree of Q to be 2.

We have

$$\text{If } X' \text{ is an independant copy of } X: \varphi_X(s)\varphi_X(-s) = \mathbb{E}[e^{is(X-X')}]$$

$$\text{But also, with } P(s) := Q(s) + Q(-s): \varphi_X(s)\varphi_X(-s) = \exp(P(s))$$

Note that P only has even terms. We write $P(s) = \sum_0^n a_k s^{2k}$.

Moreover, for if $t \in \mathbb{R}$, $\varphi_X(t)\varphi_X(-t) = |\varphi_X(t)|^2$.
Therefore $\forall t \in \mathbb{R}$, $P(t) = \log|\varphi_X(t)|^2 \in \mathbb{R}$. It follows that necessarily, the coefficients of P are all real.

Additionnally, since for real arguments t we have $|\varphi_X(t)|^2 \leq 1$, the leading coefficient of P must be negative.

Otherwise, for $t > 1$ we have :

$$\begin{aligned} \exp(P(t)) &\geq \exp(a_n t^{2n} - nA t^{n-2}) \\ &= \exp\left(t^{2n}\left(a_n - \frac{nA}{t^2}\right)\right) \xrightarrow{t \rightarrow \infty} \infty \end{aligned}$$

Where A is the magnitude of the largest of the coefficient of P .

As a result, we can write $-\gamma^2 := a_n$.

Let $N \in \mathbb{N}$, $N > 1$ and $z := \sqrt[n]{N}$. Plugging z into φ_X we get :

$$\begin{aligned} |\varphi_X(z)\varphi_X(-z)| &= |\exp(P(z))| = \left| \exp\left(\sum_{k=0}^{n-1} a_k z^{2k} + \gamma^2 N\right) \right| \\ &\geq \exp\left(\gamma^2 N - A \sum_{k=0}^{n-1} |z^{2k}|\right) \\ &\geq \exp\left(\gamma^2 N - nAN^{1-\frac{1}{n}}\right) \end{aligned}$$

Since, because $N > 1$, $|z^{2(n-1)}| = N^{1-\frac{1}{n}} \geq |z^{2k}|$

$$= \exp\left(N\left(\gamma^2 - \frac{nA}{N^{\frac{1}{n}}}\right)\right)$$

Which is our lower bound.

Now we will work to obtain our upper bound using the other expression of φ_X :

$$\begin{aligned}
|\varphi_X(z)\varphi_X(-z)| &= |\mathbb{E}[\exp(iz(X - X'))]| \\
&\leq \mathbb{E}[|\exp(iz(X - X'))|] \text{ by Jensen's inequality} \\
&= \mathbb{E}[\exp\left(-\sin\left(\frac{\pi}{2n}\right)^{2n}\sqrt{N}(X - X')\right)] \\
&\text{since } -\sin\left(\frac{\pi}{2n}\right)^{2n}\sqrt{N} \text{ is the imaginary part of } z \\
&= \varphi_X(i^{2n}\sqrt{N}s)\varphi_X(-i^{2n}\sqrt{N}s) \\
&= |\exp\left(P(i^{2n}\sqrt{N}s)\right)|
\end{aligned}$$

With $s := \sin\left(\frac{\pi}{2n}\right)$

Now let us evaluate $P(i^{2n}\sqrt{N}s)$:

$$\begin{aligned}
P(i^{2n}\sqrt{N}s) &= -(-1)^n \gamma^2 N s^{2n} + \sum_{k=0}^{n-1} (-1)^k a_k s^{2k} N^{\frac{2k}{n}} \\
&\leq \gamma^2 N s^{2n} + n A N^{1-\frac{1}{n}} = N(\gamma^2 s^{2n} + \frac{nA}{N^{\frac{1}{n}}})
\end{aligned}$$

So finally we obtain the upper bound:

$$|\varphi_X(z)\varphi_X(-z)| \leq \exp(N(\gamma^2 s^{2n} + \varepsilon_N)) \quad (4)$$

$$|\varphi_X(z)\varphi_X(-z)| \geq \exp(N(\gamma^2 - \varepsilon_N)) \quad (5)$$

With $\varepsilon_N := \frac{nA}{N^{\frac{1}{n}}}$ (note that $\varepsilon_N \xrightarrow{N \rightarrow \infty} 0$)

Recall that $s = \sin\left(\frac{\pi}{2n}\right)$, so $-1 \leq s^{2n} \leq 1$. So the two inequalities, which are true for any $N > 1$, can not both hold unless $s = \pm 1$, which means $\sin\left(\frac{\pi}{2n}\right) = \pm 1$, so necessarily $n = 1$, which means that we can write $P(t) = a_0 + a_1 t^2 = -\gamma^2 t^2$. Since we have $\varphi_X(0) = 1$, $a_0 = 0$ (it can also be seen from the fact that $X - X'$ has 0 mean).

So finally, we have $\varphi_X(t)\varphi_X(-t) = e^{-\gamma^2 t^2}$ which is the generating function of a Gaussian distribution. By Cramér's theorem, this means that X also follows a Gaussian distribution. $\square$

3.3 Cumulant method with Brillinger mixing for Point Processes

Now that we have established the basis of the cumulant method, we shall see a common hypothesis used to prove the fact that the cumulants vanish.

3.3.1 Framework

Let Φ be a stationary point-process of cumulant measure $(m^{(p)})_{p \in \mathbb{N}^*}$. We study the following statistic of Φ :

$$X_r = \sum_{x \in P} h(x/r) \quad (6)$$

Where h is assumed to be Schwartz. We can additionally assume that X_r is centered (up to subtracting the expected value). This means :

$$\mathbb{E}[X_r] = \mathbb{E}\left[\sum_{x \in \Phi} h(x/r)\right] = 0$$

Which by Campbell formula gives us:

$$\mathbb{E}[X_r] = \mathbb{E}\left[\int_{\mathbb{R}^d} h(x/r) dM(x)\right]$$

Where $M(x)$ is the mean measure of the point process. However, we have assumed Φ to be stationary, so its mean measure is equal to λLeb for some $\lambda > 0$. So it is equivalent to assume $\int_{\mathbb{R}^d} h(x) dx = 0$, which we will.

The result of interest for us it to prove the central limit theorem on the statistic X . In other words, we want to show :

$$\frac{X_r}{\sigma(r)} \xrightarrow[r \rightarrow \infty]{\mathcal{L}} \mathcal{N}(0, 1)$$

3.3.2 Usual hypotheses

The following two hypotheses are often used to show our desired central limit theorem:

1. The point process Φ is Brillinger mixing if for all $p > 2$:

$$B_p := \sup_{x \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^{p-1}} |m^p(x, [x]_2^p)| d[x]_2^{(p)} < +\infty \quad (7)$$

2. The variance $\sigma(r)$ of X_r satisfies the following :

$$\lim_{r \rightarrow \infty} \frac{\sigma^2(r)}{r^\nu} \xrightarrow[r \rightarrow \infty]{} +\infty \quad (8)$$

For some $\nu > 0$.

The first condition is tailored so that the convergence of the cumulant occurs given the second assumption. It roughly corresponds to asymptotic independance where instead of the asymptotic being a parameter n of a sequence of i.i.d. variables, it is the distance growing large. It definitely is NOT a necessary condition, as there are many cases in which the point process is not Brillinger mixing but the CLT is still verified (see for example [Gabriel Mastrili]).

3.3.3 Derivation of the Central Limit Theorem given the assumptions

To show our CLT, we will use the cumulant method: we will show that

$$\kappa^{(p)}\left(\frac{X_r}{\sigma(r)}\right) \xrightarrow[r \rightarrow \infty]{} 0$$

for $p > 2$ large enough.

Proposition 3.1. *Assumption ((7)) and ((8)) imply that X_r satisfies a CLT (given the appropriate assumptions on P and h such as stationarity of P , integrability of h and so on).*

Proof. Let some $p > 2$. We straight-forwardly evaluate $\kappa^{(p)}\left(\frac{X_r}{\sigma(r)}\right)$:

$$\begin{aligned}\kappa^{(p)}\left(\frac{X_r}{\sigma(r)}\right) &= \sum_{\gamma} (-1)^{|\gamma|} (|\gamma| - 1)! \left\langle \prod_{i=1}^{|\gamma|} \left(\frac{h}{\sigma(r)}\right)^{\otimes |\gamma_i|}, m_T^{|\gamma|} \right\rangle \\ &= \frac{1}{\sigma(r)^p} \sum_{\gamma} (-1)^{|\gamma|} (|\gamma| - 1)! \left\langle \prod_{i=1}^{|\gamma|} h^{\otimes |\gamma_i|}, m_T^{|\gamma|} \right\rangle \\ &\leq \frac{1}{\sigma(r)^p} \sum_{\gamma} (p-1)! \int_{(\mathbb{R}^d)^{|\gamma|}} \prod_{i=1}^{|\gamma|} \left| h^{\otimes |\gamma_i|} \left(\frac{x_i}{r}\right) \right| \left| m_T^{(|\gamma|)}([x]_1^{|\gamma|}) \right| d[x]_1^{|\gamma|}\end{aligned}$$

Note that $\sum_{\gamma}$ means that we are summing over all partitions of $\{1, \dots, n\}$. Here we split the first variable from the rest to use the Brillinger mixing hypothesis

$$= \frac{1}{\sigma(r)^p} \sum_{\gamma} (p-1)! \int_{(\mathbb{R}^d)^{|\gamma|}} |h^{\otimes |\gamma_1|}(\frac{x_1}{r})| \prod_{i=2}^{|\gamma|} |h^{\otimes |\gamma_i|}(\frac{x_i}{r})| |m_T^{(|\gamma|)}(x_1, [x]_2^{|\gamma|})| d[x]_1^{|\gamma|}$$

Since h is bounded by $L > 0$: $\leq \frac{1}{\sigma(r)^p} \sum_{\gamma} (p-1)! \int_{(\mathbb{R}^p)^{|\gamma|}} |h^{\otimes |\gamma_1|}(\frac{x_1}{r})| L^p |m_T^{(|\gamma|)}(x_1, [x]_2^{|\gamma|})| d[x]_1^{|\gamma|}$

$$\begin{aligned}\text{By Brillinger mixing : } &\leq \frac{1}{\sigma(r)^p} \sum_{\gamma} (p-1)! L^p B_p \int_{\mathbb{R}^d} |h^{\otimes \gamma_1}(\frac{x_1}{r})| dx_1 \\ &= \frac{1}{\sigma(r)^p} \sum_{\gamma} (p-1)! L^p B_p r^d \int_{\mathbb{R}^d} |h^{\otimes \gamma_1}(x_1)| dx_1 \\ &\leq \frac{1}{\sigma(r)^p} \sum_{\gamma} (p-1)! L^p B_p r^d L'\end{aligned}$$

So finally we get :

$$\kappa^{(p)}\left(\frac{X_r}{\sigma(r)}\right) \leq C \frac{r^d}{\sigma(r)^p}$$

Which we can rewrite as :

$$\kappa^{(p)}\left(\frac{X_r}{\sigma(r)}\right) \leq C \frac{r^d}{r^{p\nu/2}} \left(\frac{r^{\nu/2}}{\sigma(r)}\right)^p$$

For $p \geq \frac{2d}{\nu}$, the first term is bounded, and the second term goes to zero by assumption.

From this we deduce that for p large enough, the cumulant of the quantity $\frac{X_r}{\sigma(r)}$ vanish as r tends to $+\infty$, which is exactly the condition laid out previously. So we have the desired result:

$$\frac{X_r}{\sigma(r)} \xrightarrow[r \rightarrow \infty]{\mathcal{L}} \mathcal{N}(0, 1)$$

□

4 Perturbed Point Processes

We consider a stationnary point process P on $\mathbb{R}^d$. We give ourselves a perturbation field on $\mathbb{R}^d$ $(\xi_x)_x \in \mathbb{R}^d$ with the $(S_x)_{x \in \mathbb{R}^d}$ i.i.d. We don't require a priori that the perturbation have moments of any order, even a finite first order moment. In some case, instead of using a field $(\xi_x)_x \in \mathbb{R}^d$, we will use $(\xi_x)_{x \in \Phi}$ if we need to use a flow on the probability space. The perturbed point process $\tilde{\Phi}$ will be defined as :

$$\tilde{\Phi} = \sum_{x \in \Phi} \delta_{x+\xi_x}$$

Our main objective will be to prove some Central Limit Theorems for some cases of smooth statistics X_r of perturbed point process where

$$X_r = \sum_{x \in \tilde{\Phi}} h\left(\frac{x}{r}\right)$$

Our main tool will be the cumulant method using Brillinger mixing.

4.1 Brillinger Mixing of perturbed Point Processes

One of the properties preserved by perturbation is the Brillinger mixing. This means that once we have perturbed our point process Φ into $\tilde{\Phi}$, then if Φ was Brillinger mixing, so is $\tilde{\Phi}$.

Proposition 4.1. *Let P be a point process such that its correlation density function $\rho^{(p)}$ is bounded for all orders $p \in \mathbb{N}^*$. Then if P is Brillinger mixing, $\tilde{P}$ is also Brillinger mixing.*

Proof. The proof is quite straight forward: we express the correlation density function of $(\tilde{P})$, $\tilde{\rho}^{(p)}$, in terms of $\rho^{(p)}$, and then we express the factorial cumulant measure of P , $\tilde{m}_T^{(p)}$ in terms of $m_T^{(p)}$. Let f be a measurable positive function. [Add $\neq$ to the summation indices]

$$\begin{aligned} \mathbb{E}\left[\sum_{\tilde{x}_1, \dots, \tilde{x}_p}^{\neq} f([\tilde{x}]_1^p)\right] &= \mathbb{E}\left[\sum_{x_1, \dots, x_p}^{\neq} f([x]_1^p + [\xi]_1^p)\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\sum_{x_1, \dots, x_p}^{\neq} f([x]_1^p + [\xi]_1^p) | \xi\right]\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\sum_{x_1, \dots, x_p}^{\neq} f_\xi([x]_1^p) | \xi\right]\right] \end{aligned}$$

Where $f_\xi([x]_1^p) = f([x]_1^p + [\xi]_1^p)$. We write $[\xi]_1^p$ to lighten the notation but it is to be understood that each S_x corresponds to its vector x . Using the independance of ξ and Φ , and using the definition of the correlation density function, we get:

$$\mathbb{E}\left[\sum_{\tilde{x}_1, \dots, \tilde{x}_p}^{\neq} f([\tilde{x}]_1^p)\right] = \mathbb{E}_\xi\left[\int_{(\mathbb{R}^d)^p} f_\xi([x]_1^p) \rho^{(p)}([x]_1^p) dx_1 \dots dx_p\right]$$

By the change of variable $x \leftarrow x + \xi$, we get:

$$\mathbb{E} \left[\sum_{\tilde{x}_1, \dots, \tilde{x}_p}^{\neq} f([\tilde{x}]_1^p) \right] = \mathbb{E}_\xi \left[\int_{(\mathbb{R}^d)^p} f([x]_1^p) \rho^{(p)}([x]_1^p - [\xi]_1^p) dx_1 \dots dx_p \right]$$

Finally, by Fubini's theorem we get:

$$\mathbb{E} \left[\sum_{\tilde{x}_1, \dots, \tilde{x}_p} f([\tilde{x}]_1^p) \right] = \int_{(\mathbb{R}^d)^p} f([x]_1^p) \mathbb{E}[\rho^{(p)}([x]_1^p - [\xi]_1^p)] dx_1 \dots dx_p$$

By unicity of the correlation density function, we can identify $\tilde{\rho}^{(p)}$:

$$\tilde{\rho}^{(p)}([x]_1^p) = \mathbb{E} \left[\rho^{(p)}([x]_1^p - [\xi]_1^p) \right] \quad (9)$$

With this result in hand, we can link the factorial cumulant measures of the perturbed point cloud with the original one.

Recall the usual expression of the cumulant density measure:

$$\tilde{m}_T^{(p)}([x]_1^{(p)}) = \sum_{\gamma \text{ partitions of } \{1, \dots, p\}} (-1)^{|\gamma|-1} (|\gamma| - 1)! \prod_{i=1}^{|\gamma|} \tilde{\rho}^{(|\gamma_i|)}([x_{\gamma_i}]_1^{|\gamma_i|})$$

Where $[x_{\gamma_i}]_1^{|\gamma_i|} = (x_{\gamma_i(1)}, \dots, x_{\gamma_i(|\gamma_i|)})$.

We will use this relation between $\tilde{m}_T^{(p)}$ and $\tilde{\rho}^{(p)}$, as well as the previous expression, to get a relation between $\tilde{m}_T^{(p)}$ and $m_T^{(p)}$.

$$\begin{aligned} \tilde{m}_T^{(p)}([x]_1^{(p)}) &= \sum_{\gamma \text{ partitions of } \{1, \dots, p\}} (-1)^{|\gamma|-1} (|\gamma| - 1)! \prod_{i=1}^{|\gamma|} \tilde{\rho}^{(|\gamma_i|)}([x_{\gamma_i}]_1^{|\gamma_i|}) \\ &= \sum_{\gamma} (-1)^{|\gamma|-1} (|\gamma| - 1)! \prod_{i=1}^{|\gamma|} \mathbb{E}[\rho^{(|\gamma_i|)}([x_{\gamma_i}]_1^{|\gamma_i|} - [\xi_{x_{\gamma_i}}]_1^{|\gamma_i|})] \end{aligned}$$

$$\text{Using the independence of the } \xi_x = \sum_{\gamma} (-1)^{|\gamma|-1} (|\gamma| - 1)! \mathbb{E} \left[\prod_{i=1}^{|\gamma|} \rho^{(|\gamma_i|)}([x_{\gamma_i}]_1^{|\gamma_i|} - [\xi_{x_{\gamma_i}}]_1^{|\gamma_i|}) \right]$$

$$\text{Simply by linearity over a finite sum} = \mathbb{E} \left[\sum_{\gamma} (-1)^{|\gamma|-1} (|\gamma| - 1)! \prod_{i=1}^{|\gamma|} \rho^{(|\gamma_i|)}([x_{\gamma_i}]_1^{|\gamma_i|} - [\xi_{x_{\gamma_i}}]_1^{|\gamma_i|}) \right]$$

We recognize the expression of $m_T^{(p)}$: $= \mathbb{E}[m_T^{(p)}([x]_1^p - [\xi]_1^p)]$

Finally, let us prove that $\tilde{\Phi}$ is Brillinger mixing, i.e. that $\sup_{x \in \mathbb{R}^d} \tilde{B}(x) < +\infty$ where

$$\tilde{B}(x) := \int_{(\mathbb{R}^d)^{(p-1)}} |\tilde{m}_T^{(p)}(x, [x]_2^p)| dx_2 \dots dx_p.$$

Let $x \in \mathbb{R}^d$. We have :

$$\begin{aligned}
\tilde{B}(x) &= \int_{(R^d)^{(p-1)}} |\tilde{m}_T^{(p)}(x, [x]_2^p)| dx_2 \dots dx_p \\
&= \int_{(R^d)^{(p-1)}} |\mathbb{E}[m_T^{(p)}(x - \xi_x, [x]_2^p - [\xi]_2^p)]| dx_2 \dots dx_p \\
\text{By positive Fubini and Jensen: } &\leq \mathbb{E} \left[\int_{(R^d)^{(p-1)}} |m_T^{(p)}(x - \xi_x, [x]_2^p - [\xi]_2^p)| dx_2 \dots dx_p \right] \\
\text{By change of variable : } &= \mathbb{E} \left[\int_{(R^d)^{(p-1)}} |m_T^{(p)}(x - \xi_x, [x]_2^p)| dx_2 \dots dx_p \right] \\
&\leq \mathbb{E}[B] < +\infty
\end{aligned}$$

Where B is simply $\sup_{x \in \mathbb{R}^d} \int_{(R^d)^{(p-1)}} |m_T^{(p)}(x, [x]_2^p)| dx_2 \dots dx_p$ which is finite since we assumed that P is Brillinger mixing. $\square$

Now that we have established that perturbation does not impact Brillinger mixing, we will have to verify the effect of the perturbation on the second assumption of our method: does the perturbation affect the variance enough to break the assumption? [This part is not used at all actually?]

4.2 The Case of Hyperuniform Point Processes

A point process Φ is said to be hyper-uniform if it satisfies the following hypothesis:

$$\lim_{r \rightarrow \infty} \frac{\text{Var}(\#\Phi \cap B(0, r))}{|B(0, r)|} = 0 \quad (10)$$

Intuitively, this means that, although the point process might be locally disordered, it exhibits strong long-range correlation. The most simple example is that of the simple lattice $U + \mathbb{Z}^d$ where U is a random variable uniform on the basis cell of the lattice $[-\frac{1}{2}, \frac{1}{2}]^d$.

Remark. *This case is morally the opposite of Brillinger mixing: the point process exhibits long-range dependance and structure, as opposed to the long-range independance of Brillinger mixing point processes.*

It can be shown than i.i.d perturbations of hyperuniform processes remain hyperuniform [proof section 9 of S. Coste paper]. Thus another simple example of an hyperuniform point process is an i.i.d perturbed stationarized lattice $(x + \xi_x + U)_{x \in \mathbb{Z}^d}$ with ξ i.i.d and U as defined in the previous example. Moreover, Hyperuniform point processes satisfy a Central Limit Theorem under some additionnal assumptions:

Theorem 4.1. *Let Φ be an hyper-uniform point process whose covariance measure is a signed measure and such that $\text{Var}(\#\Phi \cap B(0, r)) \geq cr^{d-\alpha}$ for some $c > 0$ and $\alpha \in]0, \frac{5d}{6}]$. Then if we assume that for every $n > 2$ we have $\mathbb{E}[(\#\Phi \cap B(0, r))^n] = \mathcal{O}(r^d)$, Φ satisfies a central limit theorem:*

$$\frac{\#\Phi \cap B(0, r) - \mathbb{E}[\#\Phi \cap B(0, r)]}{\sqrt{\text{Var}(\#\Phi \cap B(0, r))}} \xrightarrow[r \rightarrow \infty]{\mathcal{L}} \mathcal{N}(0, 1)$$

Proof. We denote $X_r := \#\Phi \cap B(0, r)$ and $Y_r := \frac{X_r - \mathbb{E}[X_r]}{\sqrt{\text{Var}(X_r)}}$ and $C_n(\cdot)$ the n -th cumulant of a random variable.

Let $n \in \mathbb{N}$, $n > 2$. By definition of κ_n we have:

$$C_n(Y_r) = \frac{C_n(X_r)}{\text{Var}(X_r)^{n/2}} = \frac{\kappa_n(B_r \times \cdots \times B_r)}{\text{Var}(X_r)^{n/2}}$$

For a fixed n our first assumption gives us that there exists a constant K_n such that $\mathbb{E}[X_r^k] \leq K_n r^d$ for $k \leq n$. By the definition of cumulants, we see that there is also a constant K'_n such that $\kappa_n(B_r \times \cdots \times B_r) \leq K'_n r^d$. The second assumption gives us that $\text{Var}(X_r)^{n/2} \geq K''_n r^{n(d-\alpha)/2}$ for some other constant K''_n . $\square$

4.3 General spectral analysis of perturbed hyperuniform point processes

4.3.1 Structure factor of point processes

A useful tool to study hyperuniform point processes is spectral analysis. For stationary point processes (which is our object of study), we can define the pair-correlation function g_2 using Palm distributions:

$$g_2(X) := \frac{1}{\lambda} \mathbb{E}^0 \left[\sum_{x \in \Phi \setminus \{0\}} \mathbb{1}_{x \in X} \right], \quad X \in \mathcal{B}(\mathbb{R}^d)$$

Following this, we can define the following Correlation measure on $\mathbb{R}^d$:

$$\mathcal{C} := \delta_0 + \lambda[g_2 - \text{Leb}]$$

which is well defined on all bounded borelians (since it is a priori a signed measure). Intuitively, the difference $h := g_2 - \text{Leb}$ quantifies the difference between the current process and a totally chaotic one (i.e. Poisson): if $h > 0$, then the points are attracted to each other, and if $h < 0$ then there is a repulsion between the points. For Poisson processes, since $g_2 = \text{Leb}$, we simply have $C = \delta_0$: the points are perfectly decorrelated.

The measure $\mathcal{C}$ is useful for formulas such as the following one:

$$\text{Cov} \left(\sum_{x \in \Phi} f(x), \sum_{x \in \Phi} g(x) \right) = \lambda \int_{\mathbb{R}^d} f \otimes g(x) \mathcal{C}(dx)$$

Where $f \otimes g(x) := \int_{\mathbb{R}^d} f(y)g(y+x)dy$. From $\mathcal{C}$, we can define the *structure measure* of the point process:

The structure measure $\mathcal{S}$ of a stationary point process Φ is the Fourier transform of its correlation measure $\mathcal{C}$:

$$\mathcal{S} := 1 + \lambda \mathcal{F}[g_2 - \text{Leb}]$$

It is often called the *structure factor* in physics literature. It can give alternative expressions of the precedent formula:

$$\text{Cov} \left(\sum_{x \in \Phi} f(x), \sum_{x \in \Phi} g(x) \right) = \frac{\lambda}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}[f](k) \overline{\mathcal{F}[g](k)} \mathcal{S}(dk)$$

and

$$\text{Var}\left(\sum_{x \in \Phi} f(x)\right) = \frac{\lambda}{(2\pi)^d} \int_{\mathbb{R}^d} |\mathcal{F}[f](k)|^2 \mathcal{S}(dk)$$

This last property can be useful because if $|\mathcal{F}[f]|^2$ is in the kernel of $\mathcal{S}$, then it has zero variance and thus:

$$\sum_{x \in \Phi} f(x) = \lambda \int_{\mathbb{R}^d} f(x) dx \quad (11)$$

in a deterministic manner.

If we assume that $\mathcal{C}$ is a signed measure (i.e. that it is the difference of two positive measure, one of them being finite), it is possible to prove that a point process Φ being hyper-uniform is equivalent to $\mathcal{S}(0) = 0$ and $\mathcal{C}(\mathbb{R}^d) = 0$, and that the property of hyperuniformity is not dependant on the choice of window, so long as it is nested, convex, open and connected.

The hypothesis is crucial as we can easily find some counter-examples, where the choice of the window does change the property of hyper-uniformity. For example, $\Phi := U + \mathbb{Z}^2$, with U uniform on $[-\frac{1}{2}, \frac{1}{2}]^2$ is hyperuniform for the ball $B(0, R)$, but not for the l_∞ ball $[-R, R]^2$. [ref S Coste paper]

If $\mathcal{S}$ has a density, we denote it by S . Given the hypothesis is verified, we can assume $\lim_{k \rightarrow 0} S(k) = 0$, and we can go further and assume that the density of the structure factor vanishes like $|k|^\alpha$ around 0. We call α the hyper-uniformity exponent of the point process and such point processes are called α -hyper-uniform.

4.3.2 Central Limit Theorems for hyperuniform point processes

We can now study smooth linear statistics of hyper-uniform point processes and try to determine if they satisfy Central Limit Theorem. It can be shown (see later in our proof 4.8) that $\text{Var}(X_r) \underset{r \rightarrow \infty}{\sim} r^{d-\alpha}$. In that case, with what we have shown in section 3.3 and if the point process is Brillinger Mixing, that the statistic X_r satisfies a Central Limit Theorem. In the critical case of $d = \alpha$, there is no generality to be made. In some case there is a Central Limit Theorem (for example in the case of Ginibre point process in $\mathbb{R}^d$ where $\alpha = 2$) but it is not always true and has to be treated on a case-by-case basis. Keep in mind that this result is true if the statistic X_r uses a smooth function (Schwartz for example) but not necessarily if the function is something like an indicatrix function.

The following theorem of [Gabriel's paper] gives us a classification for the case of the indicatrix function:

Theorem 4.2. *If $h = \mathbb{1}_{B(0,r)}$ then we have the following cases:*

1. *If $\alpha > 1$, then $\text{Var}(X_r) \underset{r \rightarrow \infty}{\sim} r^{d-1}$*
2. *If $\alpha < 1$, then $\text{Var}(X_r) \underset{r \rightarrow \infty}{\sim} r^{d-\alpha}$*
3. *If $\alpha = 1$, then $\text{Var}(X_r) \underset{r \rightarrow \infty}{\sim} \log(r)$*

Remark. *It is interesting that the 'rigidity' introduced by the hyper-uniformity exponent can not overcome the intrinsic fluctuations induced by the sharp border of the indicatrix function.*

Now back to the smooth case!

If we take an α -hyperuniform point process Φ and perturb it by i.i.d. displacements ξ into $\tilde{\Phi}$, then the structure factor of the new point process $\tilde{\Phi}$ can be expressed in terms of the characteristic function of the perturbation, and that of the original structure factor:

$$\mathcal{S}_{\tilde{\Phi}} = 1 - |\varphi_S(k)|^2 + |\varphi_S(k)|^2 \mathcal{S}_{\Phi}(k)$$

If we assume that $1 - \varphi_S$ decays like $|k|^\beta$ around 0, then we have:

$$\mathcal{S}_{\tilde{\Phi}}(k) \underset{k \rightarrow 0}{\sim} c|k|^{\alpha \wedge \beta}$$

The parameter β dictates different behavior for the perturbation:

- If $\beta \in]0, 1[$, then it has no mean.
- If $\beta = 1$ then it has a mean only and no higher moments
- If $\beta \in]1, 2]$ then it has moments of order β' for $\beta' < \beta$.
- If $\beta > 2$ then the variance is 0 and then the perturbation is simply constant.

Now that we have established the behaviour around 0 of the structure element of the perturbed point process, we have [Ref paper estimating.. by Gabriel]:

$$\text{Var} \left[\sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} h\left(\frac{x}{r}\right) \right] \underset{r \rightarrow \infty}{\sim} C r^{d - \alpha \wedge \beta}$$

for some $C > 0$ and $0 < j < 1$. As a result, since necessarily $\beta \leq 2$, if $d \geq 3$ then there will be a Central Limit theorem. This means that interesting behaviour can arise for $d = 1, 2$, where the fluctuations may not be gaussians but instead could follow a stable distribution. We will focus on the case:

- $d = 1$
- $\alpha \wedge \beta > 1$
- $\beta < 2$
- $\beta < \alpha$

And we want to see if we have, in general, a "stable CLT", where the fluctuations may not be gaussians.

It is relatively easy to construct α -hyperuniform point processes by perturbing an already existing one (for example the stationarized $\mathbb{Z}^d$ lattice) by picking the right perturbation with the proper β . In that case, perturbing such a point process amounts to just a different perturbation of the underlying process. As a result we would like to prove it in the general case where the initial point process has hyperuniformity exponent α is not necessarily obtained by perturbation.

4.4 Deriving the main result

In this section we will try to show our main result: that we have a convergence (up to a renormalisation that is to be determined) of the statistic :

$$\sum_{x \in \tilde{\Phi}} h\left(\frac{x}{r}\right)$$

To a stable distribution that is not gaussian. To do so, we will look at its characteristic function and then use a "Janossy expansion" on it. However we will need to do a truncation of the point process to be able to use the expansion. Following that, we will be using a shot-noise-like approximation, followed by using our assumption on the characteristic function of ξ .

Our working assumptions for this are:

$$\begin{aligned} 1 - \varphi_\xi(t) &\underset{k \rightarrow 0}{\approx} c|k|^\beta \\ S(k) &\underset{k \rightarrow 0}{\approx} c'|k|^\alpha \\ 1 < \beta < \alpha, &\text{ with } \beta \leq 2 \end{aligned} \tag{12}$$

We also assume that h is Schwartz function with integral equal to 0. We also require an additional technical assumption on the characteristic function of the perturbation :

We need that :

$$\varphi_\xi(t) = 1 - c|k|^\beta + C|k|^{\beta+\nu} \text{ for some } \nu > 0, C > 0 \tag{13}$$

Remark. *This condition means we require $\varphi_\xi(t) = 1 - c|k|^\beta + \mathcal{O}(r^{\beta+\nu})$ which is stronger than just $\varphi_\xi(t) = 1 - c|k|^\beta + o(r^\beta)$. However it seems that we can not get away without it as we will see later.*

Our goal will be to show that the characteristic function of $\tilde{X}_r$, with proper renormalisation (which we will determine retroactively) converges to that of a stable function. Our first step to show this result will be to use the "Janossy expansion" (name ?!) [REF Paper ???] with the following functional:

$$\Psi_r(\Phi)(t) = e^{-itr^\gamma \sum_{x \in \Phi} h(\frac{x}{r})}$$

Where $\gamma \in \mathbb{R}$ for now, and we may get conditions on it later.

The "Janossy expansion" gives for an **almost surely finite point process** Φ :

$$\begin{aligned} \mathbb{E}[\Psi_r(\Phi)(t)] &= p_0 + \sum_{n \geq 1} \int_{\mathbb{R}^n} \Psi(\sum_{i=1}^n \delta_{x_i}) \Pi_n([x]_1^n) d[x]_1^n \\ &= p_0 + \sum_{n \geq 1} \int_{\mathbb{R}^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r})} \Pi_n([x]_1^n) d[x]_1^n \end{aligned}$$

4.4.1 Truncating the point process

Using the "Janossy expansion" on $\tilde{\Phi}$ is not possible directly (or Φ for that matter) because they are obviously not almost surely finite. To circumvent this issue, we will truncate our point process and only look at the following quantity :

$$\Phi_r := \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} \delta_x$$

Where $\delta > 0$ is a value that we will tune later to help us achieve, hopefully, our limit theorem to a stable distribution.

(! Notations aren't good, Ψ refers to the $e^{-it\cdot}$ and also the version without exponential !) Since Φ_r and $\tilde{\Phi}_r$ are almost surely finite and we can safely use the "Janossy expansion" on them for the functionnal $\Psi_r(\Phi_r) = r^\gamma \sum_{x \in \Phi_r} h\left(\frac{x}{r}\right)$. However, it is not clear that the part that we left out can be safely ignored when we will pass to the limit $r \rightarrow \infty$.

To do so, we will use a very similar (almost identical to be frank) as in Gabriel Mastrioli's paper [ref]: we will justify that $\Psi_r(\tilde{\Phi}_c) := \Psi_r(\tilde{\Phi}) - \Psi_r(\tilde{\Phi}_r) \xrightarrow[r \rightarrow \infty]{\mathbb{P}} 0$ by showing that $\mathbb{E} \left[\left| \Psi_r(\tilde{\Phi}_c) \right| \right] \xrightarrow[r \rightarrow \infty]{} 0$.

Proposition 4.2. *We have the following bound on $\mathbb{E} \left[\left| \Psi_r(\tilde{\Phi}_c) \right| \right]$:*

$$\mathbb{E} \left[\left| \Psi_r(\tilde{\Phi}_c) \right| \right] \lesssim \lambda r^\gamma \left(\frac{r^k}{r^{(k-1)\delta}} + \frac{1}{r^{(\beta'-1)\delta}} \right) \quad (14)$$

For $1 < \beta' < \beta$ and some $k > 1$.

Remark. *The parameters δ and γ will have to follow certain conditions in the following parts of the computation. We will come back to this part to verify that the conditions are compatible.*

Proof. We have :

$$\mathbb{E} \left[\left| \Psi_r(\tilde{\Phi}_c) \right| \right] \leq \sum_{x \in \Phi_c} r^\gamma \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \right| \right]$$

For some $x \in \Phi_c$, let :

$$\Delta = \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \right| \right]$$

We have :

$$\Delta = \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \right| \mathbb{1}_{|x+\xi| \geq \frac{|x|}{2}} \right] + \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \right| \mathbb{1}_{|x+\xi| < \frac{|x|}{2}} \right]$$

We use the fact that h is Schwartz to obtain an upper bound on the first term, and we will use Markov's inequality on the second, in combination with the fact that ξ has moments of order $\beta' < \beta$.

Let us tackle the first term:

We want to use the Schwartz character of h . As a result we write :

$$A = \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \right| \mathbb{1}_{|x+\xi| \geq \frac{|x|}{2}} \right] = \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \left(\frac{x+\xi}{r}\right)^k \left(\frac{r}{x+\xi}\right)^k \right| \mathbb{1}_{|x+\xi| \geq \frac{|x|}{2}} \right]$$

For some $k > 1$. Now we use the condition in the indicatrix function and the fact that h is Schwartz :

$$A \leq \mathbb{E} \left[\sup_{y \in \mathbb{R}} |h(y)y^k| \left| \frac{(2r)^k}{|x|^k} \right| \right]$$

And $\sup_{y \in \mathbb{R}} |h(y)y^k| < \infty$ since h is Schwartz.

Now the second term :

$$B = \mathbb{E} \left[\left| h\left(\frac{x+\xi}{r}\right) \right| \mathbb{1}_{|x+\xi| < \frac{|x|}{2}} \right]$$

Since $|x + \xi| < \frac{|x|}{2} \implies |\xi| \geq \frac{x}{2}$, we have by inequality on the indicator functions :

$$B \leq \mathbb{E} \left[\left| h \left(\frac{x + \xi}{r} \right) \right| \mathbb{1}_{|\xi| \geq \frac{|x|}{2}} \right]$$

And since h is bounded, we can simply write :

$$B \leq \mathbb{P}(|\xi| \geq \frac{x}{2}) \|h\|_\infty$$

And now by Markov's inequality with $\beta' < \beta$:

$$B \leq \|h\|_\infty \frac{\mathbb{E} \left[|\xi|^{\beta'} \right] 2^{\beta'}}{x^{\beta'}}$$

Putting together A and B we get :

$$\mathbb{E} \left[\left| \Psi_r(\tilde{\Phi}_c) \right| \right] \lesssim r^\gamma \mathbb{E} \left[\sum_{\substack{x \in \Phi \\ |x| > r^d}} \left(\frac{r^k}{|x|^k} + \frac{1}{|x|^{\beta'}} \right) \right]$$

By Campbell's formula, this gives:

$$\begin{aligned} \mathbb{E} \left[\left| \Psi_r(\tilde{\Phi}_c) \right| \right] &\lesssim r^\gamma \int_{|x| > r^\delta} \lambda \left(\frac{r^k}{|x|^k} + \frac{1}{|x|^{\beta'}} \right) dx \\ &\lesssim r^\gamma \left(\frac{r^k}{r^{(k-1)\delta}} + \frac{1}{r^{(\beta'-1)\delta}} \right) \end{aligned}$$

□

Remark. *So far, we have yet to pick our parameters γ and δ and we don't know the conditions on them. So we will leave this as is for now, and come back to it later when we have checked what conditions must γ and δ follow, and pick them accordingly.*

As a result, we will only look at the law of $\Psi_r(\tilde{\Phi}_r)$ on the truncated $\tilde{\Phi}$. If the parameters can be chosen properly, it is enough to determine the limit of the law of the statistic on the entire $\tilde{\Phi}$ as we would have $\Psi_r(\tilde{\Phi}_c) \xrightarrow[r \rightarrow \infty]{\mathcal{L}} 0$.

4.4.2 Shot-noise approximation

We will now use a "shot-noise"-like approximation to isolate the contribution of the perturbations and exploit our assumptions on φ_ξ . The approximation that we use is the following :

$$h\left(\frac{x_i + \xi_i}{r}\right) \underset{r \rightarrow \infty}{\approx} h\left(\frac{x_i}{r}\right) + h'\left(\frac{x_i}{r}\right) \frac{\xi_i}{r}$$

Which resembles a first-order Taylor approximation and draws inspiration from the paper [] by Mastrilli. We first give the computation, and we will prove that the approximation is justified after.

First, we're gonna carry out the calculation by using the approximation as if it was justified, and we will justify after that the difference between the "real" and "approximated" expressions vanishes as $r \rightarrow \infty$.

We denote $B(0, r^\delta)$ as B_r to lighten the notation slightly.

Proposition 4.3. *We have:*

$$\mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{=} p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \mathbb{E} \left[e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i}{r}\right) + h'\left(\frac{x_i}{r}\right) \frac{\xi_i}{r}} \right] \Pi_n(d[x]_1^n) + \mathcal{O}\left(r^{\gamma-1-\frac{1}{p}+\delta}\right)$$

Proof. We will first do the calculation without justifying the approximation just to show the idea, and then we justify that the approximation error is $\mathcal{O}\left(r^{\gamma-1-\frac{1}{p}+\delta}\right)$. Let us use the "Janossy expansion" on $\Psi_r(\tilde{\Phi}_r)$:

$$\mathbb{E}[\Psi_r(\tilde{\Phi}_r)] = p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \mathbb{E} \left[e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i + \xi_i}{r}\right)} \right] \Pi_n(d[x]_1^n) \quad (15)$$

Using the approximation in the previous expression yields:

$$\mathbb{E}[\Psi_r(\Phi_r)(t)] \underset{r \rightarrow \infty}{\approx} p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \mathbb{E}[e^{-itr^\gamma \sum_{i=1}^n (h(\frac{x_i}{r}) + h'(\frac{x_i}{r}) \frac{\xi_i}{r})}] \Pi_n(d[x]_1^n) \quad (16)$$

Using the independence of the ξ_i we get:

$$\mathbb{E}[\Psi_r(\Phi)(t)] = p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r})} \prod_{i=1}^n \mathbb{E}[e^{-itr^{\gamma-1} h'(\frac{x_i}{r}) \xi_i}] \Pi_n(d[x]_1^n) \quad (17)$$

There, we identify φ_ξ , the characteristic function of ξ , which finally gives us:

$$p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r})} \prod_{i=1}^n \varphi_\xi(tr^{\gamma-1} h'(\frac{x_i}{r})) \Pi_n(d[x]_1^n) \quad (18)$$

Which will allow us to use our assumption on the behavior around 0 of φ_ξ . But first, let us justify our approximation in (17).

We label as A the expression in (15) and B the expression in (17). Let us look at:

$$A - B = \sum_{n \geq 1} p_n(B_r) \int_{B(0, r^\delta)} \mathbb{E} \left[e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i + \xi_i}{r}\right)} - e^{-itr^\gamma \sum_{i=1}^n \left(h\left(\frac{x_i}{r}\right) + h'\left(\frac{x_i}{r}\right) \frac{\xi_i}{r}\right)} \right] \Pi_n(d[x]_1^n) \quad (19)$$

Using the classical inequality :

$$\left| \prod_{i=1}^n a_i - \prod_{i=1}^n b_i \right| \leq \sum_{i=1}^n |a_i - b_i| \quad (20)$$

Where $(a_i)_{i=1, \dots, n}, (b_i)_{i=1, \dots, n}$ are complex numbers of modulus 1, we get :

$$|A - B| \leq |t| r^\gamma \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \sum_{i=1}^n \mathbb{E} \left[\left| h\left(\frac{x_i + \xi_i}{r}\right) - \left(h\left(\frac{x_i}{r}\right) + h'\left(\frac{x_i}{r}\right) \frac{\xi_i}{r} \right) \right| \right] \Pi_n(d[x]_1^n) \quad (21)$$

To get a bound on this we will look at :

$$\Delta_1(x) := \left| h\left(\frac{x+\xi}{r}\right) - \left(h\left(\frac{x}{r}\right) + h'\left(\frac{x}{r}\right) \frac{\xi}{r} \right) \right|$$

Using Taylor-Lagrange formula we get:

$$\Delta_1 \leq \frac{1}{2} \left| \int_{\frac{x}{r}}^{\frac{x+\xi}{r}} \left| \frac{x}{r} - \tau \right| |h''(\tau)| d\tau \right|$$

Using Hölder for some $p > 0$ (we will determine the conditions on p later) we get:

$$\begin{aligned} \Delta_1 &\leq \frac{1}{2} \left| \int_{\frac{x}{r}}^{\frac{x+\xi}{r}} \left| \frac{x}{r} - \tau \right|^p d\tau \right|^{\frac{1}{p}} \left(\int_{\mathbb{R}} |h''(\tau)|^{\frac{p}{p-1}} d\tau \right)^{1-\frac{1}{p}} \\ &\leq \frac{1}{2(p+1)^{\frac{1}{p}}} \|h''\|_{\frac{p}{p-1}} \frac{|\xi|^{1+\frac{1}{p}}}{r^{1+\frac{1}{p}}} \end{aligned}$$

We see that if we want this quantity to have a bounded expected value, we need $1 + \frac{1}{p} < \beta$. Since the previous inequality holds for arbitrary p , we can pick it that way for now and maybe there will be additional conditions later. The norm of h'' gives us no additional constraints since h is Schwartz.

Plugging this into (21) gives us:

$$|A - B| \lesssim |t| r^{\gamma - (1 + \frac{1}{p})} \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \sum_{i=1}^n \mathbb{E} \left[\left| \xi^{1 + \frac{1}{p}} \right| \right] d\Pi_n([x]_1^n) \quad (22)$$

$$\lesssim |t| r^{\gamma - (1 + \frac{1}{p})} \sum_{n \geq 1} n p_n(B_r) \int_{B_r^n} \Pi_n(d[x]_1^n) \quad (23)$$

Since Π_n is a probability distribution, the integral simply equals 1. After that, the sum becomes simply the expected number of points inside the ball of radius r^δ , which is $2\lambda r^\delta$ by stationnarity of Φ .

Finally, we obtain the following bound:

$$|A - B| \lesssim |t| r^{\gamma - 1 - \frac{1}{p} + \delta} \quad (24)$$

□

Remark. To get a satisfactory result, we would need :

$$\gamma + \delta < 1 + \frac{1}{p} \quad (25)$$

We will continue our calculation and in the end we will regroup all the conditions that we have obtain along the way and pick our parameters so that all the conditions are met and all the quantities vanish as $r \rightarrow \infty$.

4.4.3 Behavior of the characteristic function around 0

Let us now work with :

$$\mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{\approx} p_0(B_r) + \sum_{n \geq 1} \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r})} \prod_{i=1}^n \varphi_\xi \left(tr^{\gamma-1} h'(\frac{x_i}{r}) \right) \Pi_n(d[x]_1^n) \quad (26)$$

We would like now to use our assumption on φ_ξ 's behavior around 0. For that to be valid, we need to assume :

$$\gamma < 1 \quad (27)$$

or the argument of φ_ξ will not converge to 0.

The assumption on φ_ξ is:

$$\varphi_\xi(t) \underset{t \rightarrow 0^+}{=} 1 - c|t|^\beta + |t|^{\beta+\nu} \varepsilon(t) \quad (28)$$

With $\varepsilon(t) \xrightarrow[t \rightarrow 0^+]{} 0$.

Proposition 4.4. *We have our first approximation:*

$$B = p_0(B_r) + \sum_{n \geq 1} \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r})} \prod_{i=1}^n \varphi_\xi \left(tr^{\gamma-1} h'(\frac{x_i}{r}) \right) \Pi_n(d[x]_1^n)$$

and we obtain the following result using the assumption on the behavior of φ_ξ around 0:

$$B \underset{r \rightarrow \infty}{=} C + \mathcal{O} \left(r^{(\delta+(\gamma-1)(\beta+\nu))} \right)$$

Where:

$$C = p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r}) + \sum_{i=1}^n \log \left(1 - c|t|^\beta r^{(\gamma-1)\beta} \left| h'(\frac{x_i}{r}) \right|^\beta \right)} \Pi_n(d[x]_1^n)$$

Proof. As in the previous section, we assume that the approximation is justified and carry out the calculation, and after that we go back and justify it.

Plugging the approximation into (26), we get:

$$\begin{aligned} \mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{\approx} p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \\ \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r})} \prod_{i=1}^n \left(1 - c|t|^\beta r^{(\gamma-1)\beta} \left| h'(\frac{x_i}{r}) \right|^\beta \right) \Pi_n(d[x]_1^n) \end{aligned} \quad (29)$$

Which gives us by repacking everything into an exponential (if needed, we take r large enough so that the argument inside the log is positive, which is possible since h' is bounded):

$$\begin{aligned} \mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{\approx} p_0(B_r) \\ + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h(\frac{x_i}{r}) + \sum_{i=1}^n \log \left(1 - c|t|^\beta r^{(\gamma-1)\beta} \left| h'(\frac{x_i}{r}) \right|^\beta \right)} \Pi_n(d[x]_1^n) \end{aligned}$$

Now let us justify the approximation : we denote (26) by B and (29) by C .

$$B - C = \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i}{r}\right)} \left(\prod_{i=1}^n \varphi_\xi \left(tr^{\gamma-1} h' \left(\frac{x_i}{r} \right) \right) - \prod_{i=1}^n \left(1 - c|t|^\beta r^{(\gamma-1)\beta} \left| h' \left(\frac{x_i}{r} \right) \right|^\beta \right) \right) \Pi_n(d[x]_1^n) \quad (30)$$

Using our assumption (28), in addition to (20), we get:

$$|B - C| \lesssim \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \sum_{i=1}^n \left| tr^{\gamma-1} h' \left(\frac{x_i}{r} \right) \right|^{\beta+\nu} \Pi_n(d[x]_1^n)$$

Similarly as in (22), we get:

$$|B - C| \lesssim r^{\delta+(\gamma-1)(\beta+\nu)} \quad (31)$$

□

Remark. This result gives us another condition for our parameters:

$$\delta < (1 - \gamma)(\beta + \nu) \quad (32)$$

Which at least makes sense since we have $\gamma - 1 < 0$.

Remark. This is where the assumption (28) comes into play: if we only assume a o , we get $\delta < \beta(1 - \gamma)$. And as we will see later with our choice of γ to get the right renormalisation, this will mean $\delta < 1$, which unfortunately breaks things down the line. To avoid this, we make the stronger assumption (28) which means we can pick δ such that $(\beta + \nu)(1 - \gamma) > \delta > 1$.

4.4.4 Final approximation

Now we make a final easy approximation and get our final result.

Proposition 4.5. Using our previous expression:

$$C = p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i}{r}\right) + \sum_{i=1}^n \log\left(1 - c|t|^\beta r^{(\gamma-1)\beta} \left| h' \left(\frac{x_i}{r} \right) \right|^\beta\right)} \Pi_n(d[x]_1^n)$$

We have:

$$C \underset{r \rightarrow \infty}{=} D + \mathcal{O}(r^{\delta+2(\gamma-1)\beta})$$

Where:

$$D = (B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i}{r}\right) - c|t|^\beta r^{(\gamma-1)\beta} \sum_{i=1}^n \left| h' \left(\frac{x_i}{r} \right) \right|^\beta} \Pi_n(d[x]_1^n) \quad (33)$$

Proof. The approximation we make here is simply the well known first order approximation of log:

$$\mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{\approx} p_0(B_r) + \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} e^{-itr^\gamma \sum_{i=1}^n h\left(\frac{x_i}{r}\right) - c|t|^\beta r^{(\gamma-1)\beta} \sum_{i=1}^n |h'\left(\frac{x_i}{r}\right)|^\beta} \Pi_n(d[x]_1^n) \quad (34)$$

This approximation is very easy to justify. We have:

$$\log(1+x) = x + x^2 g(x)$$

With g a bounded function around 0. Here, since $\gamma - 1 < 0$ and h' is bounded, we easily have that:

$$c|t|^\beta r^{(\gamma-1)\beta} h'\left(\frac{x}{r}\right) \xrightarrow{r \rightarrow \infty} 0.$$

As a result we get:

$$|C - D| \leq \sum_{n \geq 1} p_n(B_r) \int_{B_r^n} \left| e^{-c|t|^\beta r^{(\gamma-1)\beta} \sum_{i=1}^n |h'\left(\frac{x_i}{r}\right)|^\beta} \left| e^{\sum_{i=1}^n -c|t|^{2\beta} r^{2(\gamma-1)\beta} |h'\left(\frac{x_i}{r}\right)|^{2\beta} g\left(-c|t|^\beta r^{(\gamma-1)\beta} |h'\left(\frac{x_i}{r}\right)|^\beta\right)} - 1 \right| \Pi_n(d[x]_1^n) \right|$$

The first exponential is simply bound by 1 and the difference is bound by:

$$\left| e^{\sum_{i=1}^n -c|t|^{2\beta} r^{2(\gamma-1)\beta} |h'\left(\frac{x_i}{r}\right)|^{2\beta} g\left(-c|t|^\beta r^{(\gamma-1)\beta} |h'\left(\frac{x_i}{r}\right)|^\beta\right)} - 1 \right| \underset{r \rightarrow \infty}{\lesssim} nc|t|^\beta r^{2(\gamma-1)\beta}$$

Where ε' is simply a function that goes to 0 as $r \rightarrow \infty$. Which gives us by similar arguments as in (22):

$$|C - D| \underset{r \rightarrow \infty}{\lesssim} |t|^\beta r^{\delta+2(\gamma-1)\beta} \quad (35)$$

□

Remark. This means that we need :

$$\delta < 2(1 - \gamma)\beta \quad (36)$$

After having done this final approximation, we can repackage everything into a functional by doing a sort of "reverse Janossy expansion" and we obtain :

$$\mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{\approx} \mathbb{E} \left[\exp \left(-itr^\gamma \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} h\left(\frac{x}{r}\right) - c|t|^\beta r^{(\gamma-1)\beta} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} \left| h'\left(\frac{x}{r}\right) \right|^\beta \right) \right] \quad (37)$$

For clarity, let us write:

$$\begin{aligned} X_r &:= r^\gamma \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} h\left(\frac{x}{r}\right) \\ Y_r &:= r^{(\gamma-1)\beta} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} \left| h'\left(\frac{x}{r}\right) \right|^\beta \\ \mathbb{E}[\Psi_r(\tilde{\Phi}_r)] &\underset{r \rightarrow \infty}{\approx} \mathbb{E} \left[e^{-itX_r - c|t|^\beta Y_r} \right] \end{aligned} \quad (38)$$

4.4.5 Convergence to the stable law : Convergence in expected value

We will now study more precisely the convergence of (38) to be able to pick γ , and from there we will work out the other conditions on the parameters retroactively. We would like to study the convergence of (38). First we give the following result regarding the convergence of the means so that we can tune the parameters.

Proposition 4.6. *For $\gamma = \frac{\beta-1}{\beta}$, we have:*

$$\begin{aligned}\mathbb{E}[Y_r] &\xrightarrow[r \rightarrow \infty]{} \lambda \|h'\|_\beta^\beta \\ \mathbb{E}[X_r] &\xrightarrow[r \rightarrow \infty]{} 0\end{aligned}$$

Proof. We would like Y_r to converge to something that is not 0. By Campbell's formula we have:

$$\mathbb{E}[Y_r] = \lambda r^{(\gamma-1)\beta} \int_{B_r} \left| h' \left(\frac{x}{r} \right) \right|^\beta dx \quad (39)$$

Remark. *Again, the mean measure is simply λLeb since Φ is stationnary.*

By a change of variable we obtain :

$$\mathbb{E}[Y_r] = \lambda r^{(\gamma-1)\beta+1} \int_{B(0, r^{\delta-1})} |h'(x)|^\beta dx \quad (40)$$

We want this to converge to something that is not 0, so necessary we want:

$$(\gamma - 1)\beta + 1 = 0 \text{ and } \delta - 1 > 0 \quad (41)$$

Or simply $\gamma = \frac{\beta-1}{\beta}$. Note that this is compatible with (27) since $\beta > 1$. In this case, we have:

$$\mathbb{E}[Y_r] \xrightarrow[r \rightarrow \infty]{} \lambda \|h'\|_\beta^\beta \quad (42)$$

Remark. *It makes sense that we need $\delta > 1$ since otherwise the window doesn't grow fast enough to outpace the $\frac{1}{r}$ shrinking inside h .*

With this choice of γ , let us now look at X_r , on average still:

$$\mathbb{E}[X_r] = \lambda r^{\frac{\beta-1}{\beta}} \int_{B_r} h \left(\frac{x}{r} \right) dx \quad (43)$$

Similarly, by the same change of variable we obtain:

$$\mathbb{E}[X_r] = \lambda r^{\frac{\beta-1}{\beta}+1} \int_{B(0, r^{\delta-1})} h(x) dx \quad (44)$$

This doesn't seem to converge to 0 but recall that we have chosen h such that $\int_{\mathbb{R}} h(x) dx = 0$. As a result, we can write:

$$\int_{B(0, r^{\delta-1})} h(x) dx = 0 - \int_{|x| > r^{\delta-1}} h(x) dx$$

Using the fact that h is Schwartz, we have for $n \in \mathbb{N}, n \geq 2$:

$$\left| \int_{B(0, r^{\delta-1})} h(x) dx \right| = \left| \int_{|x| > r^{\delta-1}} \frac{x^n}{x^n} h(x) dx \right| \leq \sup_{x \in \mathbb{R}} |x^n h(x)| \left| \int_{B(0, r^{\delta-1})} \frac{1}{x^n} dx \right| \lesssim \frac{1}{r^{(n-1)}}$$

So if we pick $n > \frac{\beta-1}{\beta} + \delta$, we obtain the convergence of the average to 0. $\square$

This result helps us pick the parameter γ but if we want to get the convergence in probability this is not enough and we need to look at the variance. But first, let us look back at the parameters we have used along the various calculation and verify that there exists a proper choice that makes all approximation error vanish as $r \rightarrow \infty$.

4.4.6 Conditions on the parameters

Now that we have fixed $\gamma = \frac{\beta-1}{\beta}$, let us look at the conditions that we have fixed on our parameters and see if there are possible choices which satisfy all conditions.

Proposition 4.7. *There exists a set of valid parameters k, β', p and $\delta > 1$ such that:*

$$\mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{=} \mathbb{E} \left[e^{-itX_r - c|t|^\beta Y_r} \right] + o(1) \quad (45)$$

Proof.

$$\begin{aligned} \mathbb{E}[\Psi_r(\tilde{\Phi}_r)] \underset{r \rightarrow \infty}{=} \mathbb{E} \left[e^{-itX_r - c|t|^\beta Y_r} \right] + \mathcal{O}(r^\gamma \underbrace{\left(\frac{r^k}{r^{(k-1)\delta}} + \frac{1}{r^{(\beta'-1)\delta}} \right)}_{(14)} + \underbrace{\mathcal{O}(r^{\gamma-1-\frac{1}{p}+\delta})}_{(24)} \\ + \underbrace{\mathcal{O}(r^{(\gamma-1)(\beta+\nu)+\delta})}_{(31)} + \underbrace{\mathcal{O}(r^{2(\gamma-1)\beta+\delta})}_{(35)}) \end{aligned} \quad (46)$$

The condition we have so far is $\delta > 1$. To have the convergence of (14), we need to have:

$$\frac{\beta-1}{\beta} + k - (k-1)\delta < 0 \quad (47)$$

Since $\delta > 1$, by picking $k > \frac{\delta + \frac{\beta-1}{\beta}}{\delta-1} > 0$ we get the desired inequality. The second term means that we need:

$$\frac{\beta-1}{\beta} - (\beta'-1)\delta < 0 \quad (48)$$

Since $\beta > 1$, we have $\frac{\beta-1}{\beta} - (\beta'-1)\delta < \beta-1 - (\beta'-1)\delta$. So $\beta' > \frac{\beta-1}{\delta} + 1$ is sufficient. We still need $\beta' < \beta$ but since $\delta, \beta > 1$, $\lfloor \frac{\beta-1}{\delta} \rfloor + 1, \beta \neq \emptyset$. So we pick for example $\beta' = \frac{(\beta-1)+\delta(\beta+1)}{2\delta}$ which satisfies both conditions. Next, the convergence of (31) means that we need:

$$\delta < 1 + \nu(1 - \frac{\beta-1}{\beta}) \quad (49)$$

Which implies the condition necessary for the convergence of (35). (49) is compatible with $\delta > 1$ since $1 - \frac{\beta-1}{\beta} > 0$

Remark. *This is where the assumption (28) comes into play. If we had no made this assumption, we would need to have $\delta < 1$ which is not compatible with the rest.*

Lastly, the convergence of (24) implies:

$$\frac{\beta-1}{\beta} - 1 + \delta < \frac{1}{p} \quad (50)$$

So we need:

$$p > \frac{\beta}{(\delta-1) + (\beta-1)} > 1 \quad (51)$$

As a result, we can see that for $\gamma = \frac{\beta-1}{\beta}$, we can pick the parameter in a valid way and have the convergence of every remainders to 0, including the one resulting from the truncation, while respecting $\delta > 1$ and $\gamma < 1$. $\square$

Remark. Note that in those approximations there are two tendencies: for the truncation, we need δ to be big (at least $\delta > 1$). This makes sense intuitively since if δ is small, the convergence to the whole point cloud is too slow compared to the shrinking $h(\frac{\cdot}{r})$ which is a factor $\frac{1}{r}$ and we will be basically be dealing with something like $\sum_{x \in \Phi} h(0)$ which wouldn't make sense in our case.

Secondly, during the approximations made in the "Janossy expansion", we need δ to be not too big (here it is $\delta < 1 + \nu(1 - \frac{\beta-1}{\beta})$) otherwise the approximation errors explode when $r \rightarrow \infty$. It makes sense intuitively since the error will grow roughly as the volume of the ball $B(0, r^\delta) \times \dots$.

If we had not made the additionnal assumption on φ_ξ , we would need $\delta < 1$ for the approximations, and $\delta > 1$ for the truncation and there would be no common ground to pick δ .

4.4.7 Convergence to the stable law : Convergence of the Variance to obtain our law of large numbers

Now that we have obtained the convergence of the average of X_r (respectively Y_r) to 0 (respectively to $\lambda \|h'\|_\beta^\beta$), if we prove that the variance goes to 0, then that means we will have a convergence in **probability** which will be enough, as we will see in the last section.

To do so, we will use a slight variation of the result of Gabriel [Ref paper] which is based on spectral analysis. In this part the assumption (12) on the behaviour of the structure factor is critical. Our result follows the exact same path, but simply doesn't require that the function is C^∞ (which Gabriel probably didn't need to assume for that specific result as well since he doesn't use it anywhere it seems). Indeed, only the L_1, L_2 , bounded and fast decreasing properties are used.

Proposition 4.8 (Asymptotics of the variance of smooth linear statistics). *For f which is $L_1(\mathbb{R})$ and $L_2(\mathbb{R})$ as well as fast decreasing (i.e. $\forall n \geq 0, |x^n h(x)| \xrightarrow{x \rightarrow \infty} 0$) and bounded, $r > 1$ and $\delta > 1$, we have:*

$$\text{Var} \left[\sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} f\left(\frac{x}{r}\right) \right] \underset{r \rightarrow \infty}{=} \mathcal{O}(r^{(1-\alpha)})$$

Remark. In [Gabriel's paper], we assume that h is Schwartz. However, the proof of that results only requires at that h is L_1 and L_2 , since we use Fourier transform, and fast decreasing. The C^∞ is not important for that result. This is important because in (37), $|h'|^\beta$ is not necessarily C^∞ anymore. However, it is still bounded, fast decreasing, L_1 and L_2 so the result may apply. We also only prove the result for the variance and for $d = 1$, but it does not seem that the proof would change much for any dimension, or for the case of covariance. However, since we only use the result for the variance and $d = 1$, this is the result we will prove here.

To prove 4.8, we first use this lemma:

Lemma 4.3 (Expression for the variance of linear statistics). *With the same assumptions as in 4.8, we have:*

$$\text{Var} \left[r^{\frac{(1-\alpha)}{2}} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} f\left(\frac{x}{r}\right) \right] \underset{r \rightarrow \infty}{=} \int_{\mathbb{R}} |\mathcal{F}[f](rk)|^2 c' |k|^\alpha dk + o(1)$$

Proof. For $r > 0$, we denote $f^r = f \mathbb{1}_{[-r, r]}$. Using the properties of the structure factor, and with a change of variable $k \leftarrow rk$ we have:

$$\begin{aligned} \text{Var} \left[r^{\frac{(\alpha-1)}{2}} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} f\left(\frac{x}{r}\right) \right] &= r^{\alpha-1} \int_{\mathbb{R}} \mathcal{F}[f^{r^{\delta-1}}](rk) \overline{\mathcal{F}[f^{r^{\delta-1}}](rk)} S(k) dk \\ &= r^\alpha \int_{\mathbb{R}} |\mathcal{F}[f^{r^{\delta-1}}](k)|^2 S\left(\frac{k}{r}\right) dk \end{aligned}$$

Now we would like to replace $\mathcal{F}[f^{r^{\delta-1}}]$ by $\mathcal{F}[f]$, the non-truncated version. As a result we write:

$$\text{Var} \left[r^{\frac{(\alpha-1)}{2}} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} f\left(\frac{x}{r}\right) \right] = \int_{\mathbb{R}} |\mathcal{F}[f]|^2 S\left(\frac{k}{r}\right) dk + \Delta_1^r - \Delta_2^r \quad (52)$$

With:

$$\Delta_1^r = r^\alpha \int_{\mathbb{R}} \left(\mathcal{F}[f^{r^{\delta-1}}](k) - \mathcal{F}[f](k) \right) \overline{\mathcal{F}[f^{r^{\delta-1}}](k)} S\left(\frac{k}{r}\right) dk \quad (53)$$

$$\Delta_2^r = r^\alpha \int_{\mathbb{R}} \left(\overline{\mathcal{F}[f^{r^{\delta-1}}](k)} - \overline{\mathcal{F}[f](k)} \right) \mathcal{F}[f](k) S\left(\frac{k}{r}\right) dk \quad (54)$$

Using Cauchy-Schwartz and the the Plancherel theorem, we bound Δ_1^r :

$$\begin{aligned} |\Delta_1^r| &\leq r^\alpha \|\mathcal{F}[f] - \mathcal{F}[f^{r^{\delta-1}}]\|_2 \|\mathcal{F}[f^{r^{\delta-1}}]\|_2 \\ &\leq r^{1+\alpha} \|S\|_\infty \|f\|_2 \|f \mathbb{1}_{[-r^{\delta-1}, r^{\delta-1}]}\|_2 \end{aligned} \quad (55)$$

Since we have supposed f fast decreasing, then (55) $\rightarrow 0$ as $r \rightarrow \infty$. The same reasoning applies to Δ_2^r .

Now let's study the main term of (52) and we will try to use our assumption on the behaviour of the structure factor near 0. To do so, we write:

$$\int_{\mathbb{R}} |\mathcal{F}[f]|^2 S\left(\frac{k}{r}\right) dk = \int_{\mathbb{R}} |\mathcal{F}[f](k)|^2 c' |k|^\alpha dk + \Delta_3^r \quad (56)$$

Where:

$$\Delta_3^r = \int_{\mathbb{R}} |\mathcal{F}[f]|^2 \left(r^\alpha S\left(\frac{k}{r}\right) - c' |k|^\alpha \right) dk$$

We want to show that $\Delta_3^r \xrightarrow{r \rightarrow \infty} 0$. Let $\varepsilon > 0$. By our assumption (12), we have:

$$\exists \mu > 0, \forall k' \in \mathbb{R}, |k| \leq \mu \implies |S(k') - c' |k'|^\alpha| \leq \varepsilon |k'|^\alpha$$

Additionnaly, we have $\forall k \in \mathbb{R}, S(k)|k|^{-\alpha} < C$ for some $C > 0$.
Naturally, we split the integral into two parts:

$$|\Delta_3^r| \leq \int_{B(0, \mu r)} |\mathcal{F}[f](k)|^2 r^\alpha \left| S\left(\frac{k}{r}\right) - \left|\frac{k}{r}\right|^\alpha \right| dk \\ + \int_{\mathbb{R} \setminus B(0, \mu r)} |\mathcal{F}[f](k)|^2 r^\alpha \left| S\left(\frac{k}{r}\right) - \left|\frac{k}{r}\right|^\alpha \right| dk$$

The first term satifies $k \leq r\mu \implies \frac{k}{r} \leq \mu$, and the second can be bounded more brutally. We obtain:

$$|\Delta_3^r| \leq \varepsilon \int_{B(0, \mu r)} |\mathcal{F}[f](k)|^2 |k|^\alpha dk + \int_{\mathbb{R} \setminus B(0, \mu r)} |\mathcal{F}[f](k)|^2 |k|^\alpha (C + c') dk$$

The second integral simply goes to 0 since the integrand has a finite integral, and for the first one we obtain:

$$\limsup_{r \rightarrow \infty} |\Delta_3^r| \lesssim \varepsilon \quad (57)$$

As a result, we have:

$$\text{Var} \left[r^{\frac{(\alpha-1)}{2}} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} f\left(\frac{x}{r}\right) \right] = \int_{\mathbb{R}} |\mathcal{F}[f](k)|^2 c' |k|^\alpha dk + \Delta_1^r - \Delta_2^r + \Delta_3^r \quad (58)$$

And we have shown that $\Delta_1^r - \Delta_2^r + \Delta_3^r \underset{r \rightarrow \infty}{=} o(1)$, which concludes the proof. $\square$

This lemma trivially implies 4.8.

Armed with this, we can show that the variance of X_r and Y_r with their renormalisation goes to 0.

Let us first look at X_r :

$$\text{Var} [X_r] = \text{Var} \left[r^{\frac{\beta-1}{\beta}} \sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} h\left(\frac{x}{r}\right) \right] \underset{r \rightarrow \infty}{=} r^{2\frac{\beta-1}{\beta}} \mathcal{O}(r^{1-\alpha}) \quad (59)$$

To obtain a convergence to 0, we need $2\frac{\beta-1}{\beta} + 1 - \alpha < 0$ or $\alpha > \frac{3\beta-2}{\beta}$.

Now for Y_r , we have:

$$\text{Var} [Y_r] = \frac{1}{r^2} \text{Var} \left[\sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} \left| h'\left(\frac{x}{r}\right) \right|^\beta \right] \underset{r \rightarrow \infty}{=} \frac{1}{r^2} \mathcal{O}(r^{1-\alpha}) \quad (60)$$

So we need $-1 - \alpha < 0$ which is always verified for $\alpha > 0$.

Remark. As we can see from 1, we have something slightly stronger than our initial assumption $\alpha > \beta$. It would seem that $\frac{3\beta-2}{\beta} \geq \alpha > \beta$ doesn't work with our technique but it is not excluded that there could still be convergence to a stable law.

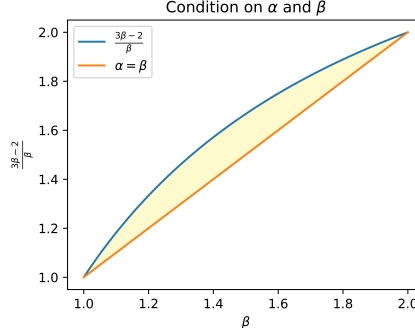


Figure 1: The condition on α so that $\text{Var}[X_r] \xrightarrow[r \rightarrow \infty]{} 0$

To recap what we have done in this part, we have obtained that:

$$X_r \xrightarrow[r \rightarrow \infty]{\mathbb{P}} 0 \quad (61)$$

$$Y_r \xrightarrow[r \rightarrow \infty]{\mathbb{P}} \lambda \|h'\|_\beta^\beta \quad (62)$$

To be able to conclude, we need to be able to pass to the limit under the expected value sign in (38) which is what we will do in our last section.

4.4.8 Convergence of the characteristic function

Proposition 4.9. *We have the following convergence:*

$$\mathbb{E} \left[e^{-itX_r - c|t|^\beta Y_r} \right] \xrightarrow[r \rightarrow \infty]{} \mathbb{E} \left[e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right] \quad (63)$$

We would like to show this using the convergence in probability of X_r and Y_r to 0 and $\lambda \|h'\|_\beta^\beta$ respectively.

Proof. Let $\varepsilon > 0$ and $A_r^\varepsilon = \{\omega, |e^{-itX_r} - 1| \leq \varepsilon\}$ and $B_r^\varepsilon = \{\omega, |e^{-c|t|^\beta Y_r} - e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta}| \leq \varepsilon\}$. We have:

$$\begin{aligned} \left| \mathbb{E} \left[e^{-itX_r - c|t|^\beta Y_r} \right] - \mathbb{E} \left[e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right] \right| &\leq \mathbb{E} \left[\mathbb{1}_{A_r^\varepsilon \cap B_r^\varepsilon} \left| e^{-itX_r - c|t|^\beta Y_r} - e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right| \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}_{(A_r^\varepsilon \cap B_r^\varepsilon)^c} \left| e^{-itX_r - c|t|^\beta Y_r} - e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right| \right] \quad (64) \end{aligned}$$

The second term can easily be bounded by $2\mathbb{P}((A_r^\varepsilon \cap B_r^\varepsilon)^c)$ since the integrand is smaller than 2. Because we have the convergence in probability, this probability goes to 0.

For the first term we write:

$$\begin{aligned} \mathbb{E} \left[\mathbb{1}_{A_r^\varepsilon \cap B_r^\varepsilon} \left| e^{-itX_r - c|t|^\beta Y_r} - e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right| \right] &\leq \\ &\mathbb{E} \left[\mathbb{1}_{A_r^\varepsilon \cap B_r^\varepsilon} \left| e^{-itX_r} \left(e^{-c|t|^\beta Y_r} - e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right) \right| \right] \\ &\quad + \mathbb{E} \left[\mathbb{1}_{A_r^\varepsilon \cap B_r^\varepsilon} \left| e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} (e^{-itX_r} - 1) \right| \right] \quad (65) \end{aligned}$$

Both of the terms are smaller than $\varepsilon \mathbb{P}(A_r^\varepsilon \cap B_r^\varepsilon) \leq \varepsilon$.

As a result, we have:

$$\limsup_{r \rightarrow \infty} \left| \mathbb{E} \left[e^{-itX_r - c|t|^\beta Y_r} \right] - \mathbb{E} \left[e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right] \right| \leq 2\varepsilon \quad (66)$$

Which concludes the proof. $\square$

Finally, what we have shown in this whole section is the following convergence:

$$\mathbb{E} \left[e^{-itr^{\frac{\beta-1}{\beta}} \sum_{|x| \leq r^\delta} h\left(\frac{x}{r}\right)} \right] \xrightarrow{r \rightarrow \infty} \mathbb{E} \left[e^{-c|t|^\beta \lambda \|h'\|_\beta^\beta} \right] \quad (67)$$

This is the convergence of the characteristic function of $r^{\frac{\beta-1}{\beta}} \sum_{|x| \leq r^\delta} h\left(\frac{x}{r}\right)$ to the characteristic function of a stable function of parameter β , and we have also shown that:

$$\sum_{\substack{x \in \Phi \\ |x| > r^\delta}} h\left(\frac{x + \xi}{r}\right) \xrightarrow[r \rightarrow \infty]{\text{distribution}} 0$$

Which means that :

$$\mathcal{L} \left(\sum_{\substack{x \in \Phi \\ |x| \leq r^\delta}} h\left(\frac{x + \xi}{r}\right) \right) \xrightarrow{r \rightarrow \infty} \mathcal{L} \left(\sum_{x \in \Phi} h\left(\frac{x + \xi}{r}\right) \right)$$

4.5 Conclusion